

TRUSTWORTHY AI

From Model Quality to System Assurance

*A qualitative engineering course from foundations
to frontier research*



◆ Current through 31 August 2026 ◆

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

ABOUT THIS EDITION

This Draft Edition 2026 is a substantial editorial and technical reconstruction of the earlier course. Its purpose is to make trustworthy AI intelligible as an engineering discipline rather than as a catalogue of desirable adjectives. The argument now progresses from a simple question - when is reliance on an AI system justified? - toward reliability, uncertainty, interpretability, fairness, privacy, security, evaluation, generative AI, agents, governance and finally continuous assurance.

The intended reader is an engineer, computer-science student, technical product leader or researcher who wants a qualitative but technically serious course. Mathematics appears only when a mathematical idea changes engineering practice; the text avoids formulae that merely make an argument look scientific. Each major concept is introduced through mechanisms, failure paths, examples, operational controls, limitations and evidence boundaries.

A fictional enterprise system called Meridian runs through the book. It begins as a retrieval-augmented research assistant, then acquires confidential data, tools, memory and limited autonomy. The system is deliberately realistic: each new capability creates an assurance problem similar to those faced by contemporary AI deployments. Real cases and current standards are used where they sharpen the argument rather than as decorative anecdotes.

The state-of-the-art discussion is current through 31 August 2026. It incorporates NIST's evolving AI Risk Management Framework, the August 2026 draft TEVV-Athlon framework, the International AI Safety Report 2026, current adversarial-ML guidance, and the EU AI Act implementation timeline as amended in 2026. Regulatory material is presented for engineering orientation and does not constitute legal advice.

The central thesis is simple: trustworthy AI is not achieved by obtaining one high score, adding an explanation screen, or passing an annual review. It is the practice of keeping system authority proportional to evidence, making assumptions visible, containing failure, and refreshing the assurance argument as the system and its environment change.

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1. Why Trustworthy AI Is an Engineering Problem

Engineering question: When is it rational to rely on an AI system, and what would justify that reliance?

Trustworthy AI becomes easier to understand when we stop treating trust as a personality trait of a model. Engineers do not ask whether a bridge is trustworthy in the abstract; they ask what loads it was designed for, how it was tested, how degradation is detected, and what happens when assumptions fail. AI requires the same change of viewpoint. The book follows Meridian, a fictional enterprise research assistant introduced as a modest retrieval-and-generation service. As Meridian acquires more users, data sources and authority, every apparently local improvement creates a new assurance question. This running case turns the subject into a sequence of engineering decisions rather than a catalogue of principles.

1.1 From Model Scores to Justified Reliance

Engineers often encounter justified reliance on AI first as a vocabulary term and only later as a design problem. Accuracy, usefulness, safety and trustworthiness are related but not interchangeable. A system can be accurate on a benchmark and still be unsafe in a workflow because the benchmark says little about users, authority, latency, data quality or consequences. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an assistant that answers 95 percent of benchmark questions correctly but invents a legal citation in the one case that triggers a contractual decision. The celebrated score and the operational failure can both be true. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

A useful engineering reformulation is to stop asking whether a model is trustworthy and ask what trustworthiness claim is being made. A strong claim contains an intended use, an operating envelope, the population or environment to which it applies, the threats and failures that are in scope, the harm that is considered unacceptable, and the evidence supporting the claim. The same model can therefore be sufficiently trustworthy for one use and plainly unsuitable for another. A summarizer for internal brainstorming and an autonomous system that approves medication should not inherit the same threshold simply because they share a foundation model.

This framing is close to the logic of safety cases in mature engineering domains. A safety case does not say that an aircraft or medical device is 'safe' in the abstract. It organizes an argument that specified hazards have been identified and controlled with evidence adequate to the expected exposure and consequence. Trustworthy AI increasingly requires the same move from adjectives to structured claims. In 2026, NIST's TEVV-Athlon draft explicitly broadens test, evaluation, verification, and validation toward real-world outcomes across statistical ML, LLMs, multimodal models, and agents. The shift is from testing a model artifact to evaluating a sociotechnical system.

Context determines the acceptable evidence threshold. A recommendation engine that occasionally proposes an irrelevant film has a very different risk profile from a system that prioritizes patients for urgent care. Even within the same domain, context matters. A radiology model used as a second reader under a trained specialist is not the same system, in assurance terms, as the identical model used autonomously in a clinic without immediate referral capacity. The model weights may be unchanged, yet the surrounding decision system, human fallback, exposure, and consequences are different.

This also changes how evidence is designed. Start with an explicit reliance decision: who may use the output, for what purpose, under which conditions, and with what fallback. Then select evidence that is directly relevant to that decision and maintain traceability between claims and tests. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. No finite evaluation proves that a learned system is correct for every future input. Even excellent empirical evidence is conditional on a distribution, interface, policy and deployment context. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

This is why deployment design belongs inside trustworthy AI rather than after it. The same probabilistic model may be safe with conservative thresholds, explicit abstention, strong monitoring, and an escalation path, but dangerous if product incentives encourage the highest possible automation rate. The relevant object of analysis is therefore the whole decision pipeline: inputs, model, retrieval or tools, user interface, policy logic, downstream action, monitoring, human role, and organizational response.

The most common conceptual error is to treat one successful test as evidence for a broader claim than the test can support. Adversarial robustness against bounded pixel perturbations is not robustness against a new hospital population. A fairness metric on a retrospective dataset is not evidence that long-run lending outcomes are fair. A mechanistic explanation of one circuit is not a proof that the model cannot exhibit an unwanted behavior elsewhere. A red-team suite is not evidence that no new jailbreak exists. A certification of an AI management system is not a certificate that every deployed model output is correct. A benchmark score is evidence, but the scope of that evidence must remain explicit.

The engineering discipline developed in this book can be summarized in one sentence: every trustworthy-AI claim should carry its own boundary conditions. The quality of a trustworthy-AI program can often be judged by how clearly it states what it does not know.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its first release is useful for summarising internal reports, yet management initially describes it as “trustworthy” because it scores well on a general QA benchmark. The engineering team replaces that adjective with narrower claims about supported document types, citation coverage and mandatory review. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

Trustworthiness is an argument, not a score



Figure 1. Claim, context, evidence and residual risk form an assurance argument.

1.2 Failure Has a Context, a Cost, and a Path to Harm

The first useful move is to make failure analysis concrete. A model error is not automatically a harm, and a harm is rarely caused by a model output alone. The engineering task is to reconstruct the path from initiating condition through system behavior, human interpretation and downstream action. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider the 2023 Mata v. Avianca sanctions episode, where fabricated legal authorities generated by ChatGPT became consequential because they were submitted without adequate verification. The lesson is not merely “LLMs hallucinate”; it is that workflow controls failed to intercept an epistemically weak artifact before it acquired institutional force. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect

answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

A practical system should prefer narrow, testable claims over grand declarations. 'The model is safe' is usually not testable. 'Under the defined document-review workflow, the system refuses to issue a final legal conclusion without at least one cited source, and every cited passage is retrievable from the approved corpus' is a meaningful claim. It can be falsified, monitored, and improved. The more an organization can transform values into observable invariants and acceptance criteria, the more trustworthy-AI work becomes engineering rather than rhetoric.

This also changes how evidence is designed. Model hazards as causal chains and annotate where prevention, detection, containment and recovery are possible. Severity and likelihood should be attached to concrete scenarios rather than to vague labels such as hallucination or bias. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Risk registers are only as good as the scenarios imagined. Unknown interactions, organisational incentives and rare combinations of conditions can remain outside the model. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, a wrong answer is initially harmless when shown as a draft. It becomes materially risky once Meridian is integrated into a procurement workflow where generated summaries influence supplier exclusion. The architecture must therefore change even though the underlying model has not. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about failure has a context, a cost, and a path to harm is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

A useful stress question for failure has a context, a cost, and a path to harm is whether the same assurance story survives a change of stakeholder. The developer wants a reproducible defect; the operator wants a stable service; the affected user wants a way to understand and challenge an outcome; the security team wants bounded authority; management wants a defensible release decision. These views are not interchangeable, but they can be connected through shared evidence and provenance. Designing that connection early reduces the common situation in which each team maintains a separate account of the system and inconsistencies are discovered only during an incident or audit.

The course therefore treats documentation as part of architecture. A requirement should point to a control; the control to a test; the test to an artifact and version; the artifact to an owner; and the owner to a review or response process. Not every link can be automated, but making the links explicit gives the organisation a map of what it actually knows. This map becomes especially valuable when systems incorporate third-party models and services whose internal development process is only partially visible.

1.3 Assurance Mindset: Claims, Evidence, and Residual Uncertainty

Engineers often encounter AI assurance first as a vocabulary term and only later as a design problem. An assurance case is a structured argument that a system is sufficiently dependable for a defined use. It connects a claim to evidence, assumptions, counter-evidence and residual uncertainty instead of compressing all confidence into a certification label. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a claim such as “Meridian does not expose confidential source text to unauthorised users.” Unit tests, access-control proofs, red-team results and audit logs can support parts of that

claim, while a known side channel or an untested connector limits its scope. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The first document should describe the decision or action the system affects. Who is affected? What can go wrong? Which harms are unacceptable? What is the maximum tolerable residual risk? What parts of the workflow can be reversed? Which users are expected to detect errors? This produces the context needed to select technical tests. Beginning with the model encourages teams to test what is easy rather than what is relevant.

Each sentence above should become a set of measurable or reviewable subclaims. The word sufficiently forces a threshold decision. The phrase under expected conditions forces an operating envelope. The phrase known shifts and attacks forces a threat model. The phrase trigger re-evaluation forces lifecycle integration. This is why assurance cases improve thinking even before they improve documentation.

Low-consequence systems may be adequately justified by retrospective evaluation, red teaming, and monitoring. High-consequence systems may require external datasets, prospective trials, formal verification of critical controls, human-factors studies, independent audit, staged deployment, and strong post-market monitoring. The MASAI mammography trial is instructive because randomized prospective evidence about workflow and detection is fundamentally stronger than a retrospective accuracy table. The evidence burden should scale with exposure and irreversibility.

This also changes how evidence is designed. Write assurance claims in falsifiable language, maintain an evidence register, record evidence freshness, identify the owner of each claim and define what system changes require re-evaluation. Treat disagreement and failed tests as useful updates to the argument. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Assurance cannot eliminate uncertainty. It can reveal where uncertainty remains, make acceptance decisions explicit and prevent confidence from silently outliving the evidence that created it. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, the team creates its first assurance ledger before adding new capabilities. The ledger is deliberately small: a handful of claims about data access, citation behavior, fallback and logging. It becomes the backbone to which later chapters add evidence. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about assurance mindset: claims, evidence, and residual uncertainty is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

For a design review, assurance mindset: claims, evidence, and residual uncertainty can be turned into a short evidence exercise. Start with one concrete claim the team expects to make at release, then list three plausible observations that would make the claim false or materially weaker. Next, ask which of those observations can be generated before deployment, which require production monitoring, and which depend on information the organisation does not currently collect. The exercise is intentionally asymmetric: it is easier to accumulate confirmatory examples than to search for conditions that would change a decision. Designing the search for disconfirmation makes testing more realistic and creates explicit triggers for escalation or rollback.

For students and practitioners, this suggests a useful habit: write the decision rule beside the metric. If a score deteriorates by five percent, what happens? If a novel input is detected, who sees it? If two evaluators disagree, is the output blocked, sampled for review, or accepted? If the answer is 'nothing', the measurement is primarily descriptive. Trustworthy engineering begins when

evidence participates in control flow and when the organisation can explain why a particular amount of residual uncertainty was acceptable for the authority that was granted.

Trustworthiness is an argument, not a score



Figure 2. Claim, context, evidence and residual risk form an assurance argument.

The important shift is now in place: trustworthiness is not a model adjective but a system argument. To make that argument meaningful, the next chapter defines what the system actually is, where its boundaries lie, and how those boundaries change over time.

2. Define the System Before You Measure It

Engineering question: What exactly is the object being evaluated when an AI product includes data, models, tools, interfaces, people and organisational rules?

Most failed assurance programs begin too late, after a team has already decided that “the model” is the unit of analysis. Real products are layered socio-technical systems. The same foundation model can be benign in a sandbox and dangerous when connected to email, a payment API or a clinical workflow. Meridian now provides a useful boundary exercise: its model is only one component among retrieval services, permissions, prompts, caches, human approvals and operational policies. Before measuring reliability or fairness, engineers must decide which of those components belong to the assurance object.

2.1 The Four-Layer System: Data, Model, Harness, Organization

A course on system decomposition should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. A practical decomposition separates data, model, harness and organization. Data supplies observations and knowledge; the model transforms representations; the harness provides retrieval, tools, memory, policies and orchestration; the organization supplies goals, incentives, roles and escalation paths. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a retrieval-augmented assistant that appears identical at the chat surface but changes behavior when a new vector index, system prompt, access-control library or review policy is deployed. Treating the neural model as the whole system misses most of the variables that govern real outcomes. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Failures propagate across these layers. A model may hallucinate because it lacks knowledge, but the harmful event may occur because the system did not retrieve an authoritative source, the interface presented the answer as definitive, the user had no reason to doubt it, and the organization had no monitoring for incorrect policy advice. The Air Canada chatbot case became famous because it dramatized a simple point: an organization remains responsible for information delivered through its AI interface. The chatbot is not an independent legal actor that absorbs liability.

At the model layer, the first trust problem is objective mismatch. Supervised learning minimizes error on labels that approximate a real-world goal. Reinforcement learning maximizes a reward that approximates desired behavior. Language-model pretraining predicts tokens even though the product goal may be truthful assistance, scientific reasoning, code generation, or safe action. Post-training reduces some gaps, but it does not erase the basic fact that training objectives are proxies for what users actually care about.

Proxy problems become more severe as systems gain autonomy. A customer-support agent rewarded for resolution rate may learn to close tickets aggressively. A coding agent rewarded for passing tests may exploit weaknesses in the tests. A research agent rewarded for novelty may produce unsupported claims that look novel. The trustworthy-AI question is not merely whether the objective correlates with the goal in the training distribution; it is how the system behaves when optimization discovers paths that satisfy the proxy while violating the intent.

This also changes how evidence is designed. Draw an architecture whose boxes correspond to trust boundaries and decision points, not vendor products. For every layer, record inputs, outputs, state, authority, dependencies, logging and failure containment. Tests can then be assigned to the component that can actually fail. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

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The boundary of the claim must remain visible. Layering is an analytical convenience rather than a perfect ontology. Some controls span layers, and organisational behavior can alter technical operation in ways that diagrams cannot fully capture. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its apparent “knowledge” is partly a property of the document corpus, partly the language model, partly retrieval ranking and partly the policy that decides whether unsupported claims may be shown. The team learns to stop filing every defect under “model quality”. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design

rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about the four-layer system: data, model, harness, organization is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The deployed object is a system



Figure 3. A deployed AI product spans data, model, harness and organisation.

Concept	Engineering meaning
Data	Training, retrieval, labels, provenance, permissions, quality and shift.
Model	Learned transformation, capabilities, limitations, uncertainty and internal behavior.
Harness	Prompts, retrieval, tools, memory, policy gates, orchestration, logging and fallbacks.
Organization	Users, incentives, owners, review procedures, escalation and accepted residual risk.

2.2 Lifecycle, Change, and Evidence Decay

The first useful move is to make lifecycle assurance concrete. AI evidence decays because the evaluated object changes. Model versions, prompts, datasets, user populations, tools and upstream APIs can shift independently. A result that was valid at release can become historical evidence rather than current evidence without any dramatic version-number change. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a safety evaluation conducted before a retrieval index is re-embedded with a new model. The generator is unchanged, yet retrieval neighborhoods change, altering factuality, sensitive-data exposure and attack surface. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Trustworthiness evidence has a half-life. Data distributions drift, models are updated, prompts change, tools gain new permissions, organizational policy changes, and attackers adapt. The correct lifecycle is therefore recursive rather than linear. Design creates claims. Training produces an artifact. Evaluation tests those claims. Deployment creates new evidence. Monitoring identifies deviations. Changes trigger re-evaluation. Incidents modify the threat model. Governance decides whether the evidence is still sufficient.

This loop is more than process hygiene. It changes what engineers optimize. If the evaluation suite cannot be rerun after a model update, it is not an assurance mechanism; it is a one-time experiment. If data provenance is unavailable, a privacy or bias finding cannot be traced. If a system update changes the retrieval corpus without recording the version, earlier factuality results may no longer apply. Trustworthiness depends on configuration management as much as on machine learning.

An assurance case has a configuration scope. A new base model, a changed system prompt, a new embedding model, a modified retrieval corpus, a new tool, a new permission, or a new user population can invalidate part of the evidence. Change-impact analysis should map each change to the claims it can affect and the tests that must be rerun. This is the bridge from static compliance to continuous assurance.

This also changes how evidence is designed. Version every assurance-relevant artifact, create change classes, and define re-test triggers. Small configuration changes can run narrow regression suites; authority-changing integrations should trigger broader review. Production telemetry should be linked back to the exact system configuration that produced it. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Change-control discipline cannot guarantee that two versions behave similarly. Learned components can exhibit nonlinear changes, and third-party services may change without transparent versioning. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Foundation-model APIs can change even when the application code remains stable. A new model snapshot may improve coding and worsen a particular refusal behavior; a new safety policy may change outputs; a new context window may alter retrieval dynamics. Version pinning, release qualification, regression suites, and change notices are therefore part of trustworthy deployment. When version pinning is impossible, the application must assume that the underlying component is dynamic and monitor it accordingly.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, each model or connector update creates an evidence-diff: which previous claims remain supported, which evidence is stale, and which tests must run again. This prevents an annual governance review from becoming a ceremony detached from the deployed system. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about lifecycle, change, and evidence decay is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

2.3 Requirements, Traceability, and the Operating Envelope

A course on AI requirements engineering should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. A useful AI requirement combines desired behavior with context and tolerance. “Be accurate” is not testable enough; “for approved internal policy documents, answer factual questions with verifiable source spans or abstain when evidence is

insufficient” creates an observable contract. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an image classifier whose average error is acceptable but whose operating environment includes night-time cameras never represented during validation. The key defect is not merely lower accuracy; it is that the deployment escaped the validated envelope. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

This also changes how evidence is designed. Define the intended users, tasks, input classes, environmental conditions, authority and prohibited uses. Link each requirement to one or more tests and to monitoring signals that can detect departure from the envelope. Make non-goals explicit so that product expansion cannot silently redefine success. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Operating envelopes are approximations of an open world. They require monitoring because some relevant variables are hard to observe, and users will discover use cases the designers did not anticipate. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its supported envelope initially excludes external legal advice and autonomous outbound communication. The interface states these limits, policy gates enforce some of them, and telemetry is designed to reveal repeated attempts to use Meridian outside the declared envelope. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small

product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about requirements, traceability, and the operating envelope is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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The final perspective on requirements, traceability, and the operating envelope is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

Once the system boundary and operating envelope are explicit, reliability becomes a much sharper question. We can now ask not whether Meridian is reliable in general, but how its behavior changes as inputs, environments and adversaries move away from the conditions under which evidence was collected.

3. Reliability in a Changing and Adversarial World

Engineering question: How should engineers reason about systems that work on yesterday's test set but encounter tomorrow's distribution?

Classical software can fail because a programmer encoded the wrong rule. Learned systems add another failure mode: the rule itself is inferred from data whose relationship to deployment is imperfect and unstable. Reliability therefore has a temporal dimension. Meridian's early success encourages teams to feed it new document types and connect it to new departments. Nothing "breaks" in the conventional sense, yet the distribution of inputs drifts away from the one used to establish confidence. The chapter develops reliability as a closed loop of envelope definition, stress, observation and repair.

3.1 Distribution Shift: IID Is a Convenience, Not a Law

A course on distribution shift should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. The independent-and-identically-distributed assumption is useful for statistical analysis but weak as a deployment story. Covariate shift, label shift, concept drift, subgroup shift and domain shift affect different parts of the data-generating process and call for different responses. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a fraud model trained during one economic regime, or a language system evaluated mostly on polished English while real users submit multilingual fragments, scans and organisation-specific jargon. Aggregate validation accuracy can remain high while a newly important slice deteriorates. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

WILDS helped make natural distribution shift a first-class benchmark problem by collecting real changes across institutions, geography, time, and acquisition conditions. Its lesson is uncomfortable: models that look strong on identically distributed test data can degrade materially when deployed into realistic new domains. This is not a corner case. Most long-lived systems eventually encounter shift because populations, behavior, hardware, policy, and competitors change.

External validation in medicine provides a particularly clear example. The original widely deployed Epic Sepsis Model performed substantially worse in an independent 2021 evaluation than many users expected from vendor-reported evidence. A 2026 prospective multicenter evaluation of the updated version found much stronger discrimination, but performance still varied by institution and the positive predictive value remained constrained by the clinical setting and alert prevalence. The engineering lesson is not that sepsis prediction is impossible. It is that transportability is an empirical claim that must be re-established locally.

Suppose a model ranks high-risk patients well but the event is rare. A threshold chosen to maximize sensitivity can generate a large number of false alerts. Those alerts consume attention, change clinician behavior, and may create alert fatigue. The model can therefore improve statistically while the operational system becomes worse. Trustworthy evaluation must connect model metrics to workflow economics: who receives an alert, how often, what action follows, what opportunity cost is created, and what happens when the model is wrong.

This also changes how evidence is designed. Establish reference distributions, slice metrics by meaningful operational variables, test plausible shifts before deployment and monitor feature, embedding, outcome and behavioral drift after deployment. Pair drift detection with a decision: investigate, reduce authority, retrain or temporarily fall back. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Drift metrics do not automatically tell whether a change is harmful. Some distribution changes are benign, and harmful concept changes can occur without obvious feature drift. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Teams often deploy drift dashboards that compare feature distributions with a training baseline. This can be useful, but a statistical change in inputs does not necessarily imply a harmful change in performance, and harmful concept drift can occur without a dramatic change in marginal feature distributions. Engineers should distinguish data drift, where observed inputs change; label or outcome drift, where the target prevalence changes; and concept drift, where the relationship between input and target changes. Only the last two directly threaten the validity of the learned mapping, but all three can be operational signals.

A mature monitor therefore combines several layers. It watches input quality and missingness, compares representation or feature distributions, tracks prediction and confidence distributions, and, when delayed ground truth becomes available, estimates real performance. For language systems, analogous signals include retrieval hit rates, citation support, refusal rates, disagreement among validators, tool failures, and the emergence of novel prompt clusters. Drift detection should trigger investigation, not automatic retraining. Retraining on contaminated or strategically manipulated data can make the problem worse.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority.

This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, usage expands from research reports to meeting transcripts and scanned contracts. Citation quality falls only on OCR-heavy material. A single overall factuality score masks the degradation, while slice-based monitoring exposes that the operating envelope has expanded without evidence. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about distribution shift: IID is a convenience, not a law is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

3.2 Robustness, Adversarial Pressure, and Failure Budgets

A course on robustness should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Robustness concerns behavior under perturbation, but perturbations must represent a threat or variability model. Random noise, semantic paraphrase, corrupted sensors, adversarial examples and prompt attacks are not interchangeable tests. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an image model resilient to Gaussian noise but vulnerable to carefully constructed adversarial inputs, or an LLM stable under paraphrase yet easily induced to follow malicious instructions embedded in retrieved content. Passing the first test says almost nothing about the second. What makes the example useful is the separation between mechanism and symptom. The

visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

These categories should not be conflated. The adversarial-example literature demonstrated that neural networks can be highly sensitive to carefully chosen perturbations that humans barely notice. Goodfellow, Shlens, and Szegedy provided an early explanation based on local linear behavior, and Madry and colleagues turned adversarial training into a robust-optimization problem. Randomized smoothing later enabled certificates for specific bounded perturbations. These results are important, but the guarantee is explicitly conditional on a perturbation model. It says little about a hospital acquiring a new scanner, a language model receiving an unfamiliar policy document, or an agent encountering a website redesigned overnight.

For deployment, it is useful to think in terms of a robustness budget. How far can inputs, environment, or components deviate before the system's risk exceeds an acceptable level? Some deviations can be detected and trigger abstention. Others should trigger a fail-safe mode. Still others require retraining or redesign. This framing encourages engineers to define operational design domains for AI, analogous to safety-critical automation: the system should know, as far as practical, when it has left the conditions under which its evidence applies.

Adversarial training remains one of the strongest empirical approaches within explicit threat models, but it can be computationally expensive and may trade clean accuracy for robustness. Data augmentation can improve invariance but only for transformations represented during training. Distributionally robust optimization focuses learning on worst-performing groups or environments but depends on how those groups are specified. Test-time adaptation can help under shift but introduces a new online learning mechanism whose own stability must be evaluated. Certified methods offer stronger guarantees but usually for narrow input regions and model classes. There is no general-purpose robustness switch.

This also changes how evidence is designed. Define expected environmental variation and attacker capabilities, then construct stress suites around them. Track not only failure rate but severity, detectability and recovery. Use failure budgets to decide which error classes can be tolerated and which require architectural containment. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Robustness testing samples an enormous space. Adaptive adversaries can search specifically for regions left uncovered by a fixed red-team suite, so repeated and diversified testing remains necessary. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but

what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

The practical conclusion is to combine methods. Train for known perturbations, validate on naturally shifted data, use anomaly and out-of-distribution signals as warning mechanisms rather than oracles, implement safe fallback behavior, and continuously compare live performance with the assumptions used during validation. Robustness is strongest when the system architecture makes graceful degradation possible.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its evaluation separates benign formatting variation from malicious document instructions. A low-severity formatting failure consumes a quality budget; a policy-bypass that could trigger a privileged tool is treated as a containment failure requiring a release block. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about robustness, adversarial pressure, and failure budgets is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Reliability is a closed engineering loop



Figure 4. Reliability is maintained through an iterative stress-observe-repair loop.

3.3 Fault Injection, Recovery, and Graceful Degradation

A course on graceful degradation should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Reliable systems are not those that never fail but those whose failures remain bounded and observable. AI pipelines depend on networks, retrieval stores, external APIs and probabilistic components, so recovery behavior deserves first-class testing. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a retrieval service returning stale results, an embedding endpoint timing out, or a tool API supplying malformed data. A naive assistant may improvise around the missing dependency and produce fluent nonsense, whereas a robust system recognises degraded evidence and reduces confidence or authority. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Traditional reliability engineering deliberately breaks components to learn how systems fail. AI systems benefit from the same attitude. Disable retrieval. Return stale documents. Corrupt a sensor. Delay a tool. Remove a required field. Make the API return partial output. Inject contradictory instructions. Replace a human approver with an unavailable service. The objective is to discover whether the system fails loudly and safely or silently and persuasively. This kind of fault injection is particularly valuable because many production incidents arise from integration behavior rather than from the isolated model.

The final reliability metric should therefore include recovery. Mean performance during perfect operation is less informative than the system's ability to detect degraded conditions, enter a safe mode, preserve state, explain what failed, and resume without duplicating harmful actions. Graceful degradation is a design property, not something that emerges automatically from a more accurate model.

This also changes how evidence is designed. Inject dependency failures, latency, stale caches, partial documents and permission errors. Define safe degraded modes such as retrieval-only output, explicit uncertainty, read-only operation or human escalation. Test recovery and replay so

that incident analysis is possible after the service returns. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Graceful degradation can reduce harm but also reduce utility. If fallback is too conservative, users route around the system; if it is too permissive, the fallback becomes an untested second product. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its tool layer is deliberately fault-injected in staging. When retrieval becomes unavailable, Meridian is forbidden from presenting unsupported enterprise facts as if grounded; it switches to a clearly marked general-model mode or abstains depending on the task. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about fault injection, recovery, and graceful degradation is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The final perspective on fault injection, recovery, and graceful degradation is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

Reliability tells us that behavior varies with conditions, but engineers still need a way to represent uncertainty inside a particular decision. The next chapter separates probability, confidence, calibration and abstention, showing why a system that knows when not to act can be more trustworthy than one that merely improves average accuracy.

4. Uncertainty, Calibration, and Safe Decisions

Engineering question: How should uncertainty change what an AI system is allowed to say or do?

An accuracy score describes a population of predictions; an operational decision occurs one case at a time. This gap is where uncertainty becomes useful. Engineers need to know not merely how often a system is correct, but whether its confidence is meaningful, whether unfamiliar cases can be detected, and whether low-confidence cases can be routed elsewhere. Meridian now faces questions whose evidence quality varies dramatically. Rather than forcing every query through the same path, the team begins to treat uncertainty as an input to control flow.

4.1 Accuracy, Confidence, and Calibration

A course on calibration should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Confidence is useful only if it corresponds to empirical reliability under relevant conditions. A classifier that assigns 0.9 probability should be correct roughly 90 percent of the time among comparable 0.9 cases if the probability is to support decisions. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a medical triage model with excellent ranking performance but systematically overconfident probabilities. The ordering of cases can be good while threshold-based decisions are unsafe because the numerical scores do not mean what users assume they mean. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Accuracy asks how often a system is correct. Calibration asks whether its stated confidence corresponds to how often it is correct. A model can be accurate but overconfident, which is dangerous when downstream software treats confidence as a risk signal. A model can also be underconfident, causing unnecessary escalation and reducing automation value. For high-stakes deployment, the goal is usually not maximum confidence but useful uncertainty that changes action appropriately.

Textbooks often separate aleatoric uncertainty, attributed to irreducible noise in the data-generating process, from epistemic uncertainty, attributed to limited knowledge or model uncertainty. The distinction is conceptually useful because the remedies differ: more data can sometimes reduce epistemic uncertainty, while irreducible ambiguity requires broader prediction sets or different decisions. In practice the two are not directly observable and can be entangled with model misspecification. Engineers should use the distinction to reason about intervention, not to claim that a deployed system has uniquely decomposed uncertainty into two true physical quantities.

Expected Calibration Error is popular because it is easy to explain, but it depends on binning choices and can hide important structure. Negative log-likelihood and the Brier score are proper scoring rules: they reward probabilistic predictions that reflect the true conditional distribution, at

least under the assumptions of the task. Reliability diagrams show where confidence departs from observed accuracy. For selective systems, risk-coverage curves reveal whether abstention actually reduces errors. In safety-relevant applications, several views should be reported together because no single scalar captures the decision problem.

This also changes how evidence is designed. Measure calibration on held-out and shifted data, inspect reliability by subgroup and confidence range, and apply post-hoc calibration only when its assumptions are credible. For language systems, avoid pretending token probability is automatically a probability that the semantic answer is true. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Calibration is distribution-dependent and can deteriorate under shift. A perfectly calibrated aggregate can also conceal badly calibrated subgroups, and calibration does not reveal whether the target itself is appropriate. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Imagine a triage classifier used to prioritize 10,000 cases per month. At a permissive threshold it automates 90 percent of cases with a 4 percent error rate. At a conservative threshold it automates 60 percent with a 1 percent error rate. Which system is better cannot be answered from model quality alone. The organization must price the cost of manual review against the consequence of error and the capacity of the human queue. The important engineering artifact is the policy that maps confidence and context to action, together with evidence that the policy remains calibrated after deployment.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its team stops displaying a synthetic “92% confidence” generated by the language model. Instead it derives evidence features from retrieval coverage, source agreement and task-specific validation, then measures whether these features predict actual answer quality. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture

and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about accuracy, confidence, and calibration is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Uncertainty matters when it changes action



Figure 5. Uncertainty is valuable when it changes orchestration and authority.

4.2 Abstention, Selective Prediction, and Conformal Guarantees

The first useful move is to make abstention and selective prediction concrete. A trustworthy system does not need to answer every case. Selective prediction trades coverage for lower error on the accepted subset; conformal methods can provide distributional coverage guarantees under explicit assumptions and are valuable precisely because those assumptions can be stated. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a document classifier that routes ambiguous cases to a human rather than forcing a label. If automatic coverage falls from 100 to 80 percent while serious error falls sharply, the

system may be more useful despite lower apparent autonomy. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Abstention converts uncertainty from a number into a control decision. Instead of forcing a prediction on every input, the system routes uncertain cases to a human, a slower model, a retrieval process, a formal checker, or a safe default. Selective prediction should therefore be evaluated with a risk-coverage curve: as the system handles fewer cases autonomously, does the error rate fall in the cases it keeps? This metric directly connects confidence with operational design.

Deep ensembles remain a strong practical baseline because independently trained models expose some epistemic variation and often improve both predictive performance and uncertainty estimates. They are expensive, but the engineering idea is simple: disagreement among plausible models contains useful information. Bayesian approximations aim at a similar goal with different computational trade-offs.

Conformal prediction is attractive because it can give finite-sample coverage guarantees under assumptions that are weaker than a fully specified probabilistic model. Instead of outputting one label or interval, a conformal method can produce a set designed to contain the correct answer at a chosen frequency. The guarantee is not magic. Exchangeability or related assumptions matter, and distribution shift can break coverage. Recent work extends conformal approaches to heterogeneous and federated settings, but engineers must state the assumptions with the guarantee.

This also changes how evidence is designed. Define an abstention policy before tuning thresholds, quantify both coverage and residual error, and include the cost and capacity of the fallback path. Where conformal prediction is applicable, report the guarantee, calibration set and exchangeability assumptions in plain engineering language. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Abstention can fail socially if users interpret it as incompetence or bypass it under workload pressure. Conformal guarantees can also break under distribution shift or poorly matched nonconformity scores. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, it learns to respond “the available documents do not support a reliable answer” for some procurement questions and automatically opens a review task. The system’s product metric becomes successful resolution rather than percentage of questions answered by the model alone. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about abstention, selective prediction, and conformal guarantees is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

4.3 Uncertainty in Language Models, Tool Use, and Decisions

The first useful move is to make uncertainty in generative systems concrete. Generative models do not expose a single well-behaved uncertainty variable for statements, plans or tool calls. Engineers therefore combine imperfect signals: self-consistency, retrieval support, verifier results, model disagreement, novelty, tool feedback and task-specific constraints. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a generated answer whose individual tokens are high probability but whose central factual claim is unsupported, or a low-probability coding suggestion that is nevertheless correct after compilation and tests. Surface likelihood and semantic correctness are different objects. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

For open-ended generation, a token probability is not the probability that an answer is true. A language model can assign high likelihood to a fluent continuation that contains a false claim, and different phrasings can express the same semantic answer. Useful uncertainty signals therefore include self-consistency across sampled answers, semantic clustering, retrieval support, verifier agreement, model ensembles, conformalized answer sets for constrained tasks, and calibrated abstention policies. None should be presented as a universal probability of truth.

The strongest design pattern for factual systems is to separate answer generation from evidence assessment. An answer that cannot be grounded in an authoritative source can be marked uncertain, withheld, or routed for review. This is more defensible than asking a generative model to introspectively report how sure it feels.

This also changes how evidence is designed. Move uncertainty estimation to the level where the decision is made. For factual answers, score claims and evidence; for code, execute tests; for tool calls, validate arguments and expected effects; for plans, check constraints and require confirmation before irreversible actions. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. No current uncertainty method turns an LLM into a calibrated oracle about its own correctness. Signals are correlated with quality in some settings and can be gamed or fail under novel conditions. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, uncertainty becomes an orchestration variable. Meridian may retrieve more evidence, call a second verifier, ask the user a clarifying question, abstain, or request approval depending on the kind of uncertainty rather than on a single confidence threshold. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about uncertainty in language models, tool use, and decisions is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The final perspective on uncertainty in language models, tool use, and decisions is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

Once uncertainty influences action, explanations acquire a clearer purpose. The question is no longer whether a model can produce an attractive rationale, but whether an explanation helps a particular actor test, contest, debug or constrain the system.

5. Explainability, Interpretability, and Causal Understanding

Engineering question: What kind of explanation can legitimately support debugging, oversight or assurance?

Explainability is often discussed as if every stakeholder needs the same thing: a picture of feature importance or a paragraph explaining the model's answer. In practice, users, developers, auditors and researchers ask different causal and procedural questions. Meridian makes the distinction visible. A procurement specialist may need to know which policy passages support an answer; an engineer may need to know why retrieval failed; an auditor may need reproducible evidence that a control operates as claimed. This chapter treats explanation as an instrument whose validity depends on its use.

5.1 Explanation for Whom, and for What Decision?

The first useful move is to make explanation objectives concrete. Explanations serve distinct purposes: justification to a user, diagnosis for an engineer, contestability for an affected person, evidence for an auditor, or scientific understanding of an internal mechanism. A method suitable for one purpose can be misleading for another. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a feature attribution that helps a developer notice that a model relies on postcode, yet cannot by itself justify an individual credit decision or prove that removing the feature would remove discriminatory effects. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

LIME, SHAP, integrated gradients, saliency maps, counterfactual explanations, prototypes, and concept-based methods all attempt to translate a complex decision into a simpler representation. They can be extremely useful for debugging and communication. Their danger is semantic overreach. A feature attribution can say that changing a feature would be associated with a change in output under the explanation method. It does not automatically show that the feature is a causal mechanism inside the model, nor that the explanation will remain stable for a nearby input.

The SHAP framework unified a large family of additive feature-attribution techniques around Shapley values, giving a principled way to distribute credit among features. Later work has shown that Shapley-based explanations still have important information-theoretic and practical limitations. The relevant question is therefore not whether a method is mathematically elegant but whether it is faithful enough for the decision being made from it.

Post-hoc explainability attracts attention because it can be applied after a complex model has already been trained. For some high-stakes structured tasks, however, the first question should be whether the black box is necessary. Generalized additive models, monotonic models, sparse rule

systems, decision sets, prototypes, and concept-bottleneck architectures can make important relationships visible by construction. Their advantage is not moral purity; it is that the explanation is more tightly coupled to the computation. Their disadvantage is that they may not reach the same performance or flexibility on highly complex data.

This also changes how evidence is designed. Start from the decision the explanation must support and define acceptance criteria such as fidelity, stability, completeness, comprehensibility and actionability. Prefer explanations tied to observable evidence, counterfactual tests or documented system traces over fluent post-hoc stories. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Interpretability always compresses. Even a faithful local explanation may omit interactions outside the local region, while an understandable global account may simplify a model beyond operational usefulness. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

The engineering decision should compare the marginal value of a more complex model with the assurance cost it creates. If a transparent model loses a fraction of a percentage point in predictive performance but enables stable audit, better debugging, and safer recourse, the system-level trade may be favorable. Conversely, using an interpretable model merely to satisfy a documentation requirement while it materially underperforms can also create harm. Interpretability is part of system optimization, not an absolute rule.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, user-facing explanations cite source passages and disclose uncertainty; developer diagnostics expose retrieval ranks, policy decisions and tool traces. Meridian deliberately presents different evidence to different actors rather than pretending one explanation satisfies all of them. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about explanation for whom, and for what decision? is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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From explanation to assurance



Figure 6. Explanation becomes stronger as claims are tested by behavior and intervention.

5.2 From Behavioral Explanation to Mechanistic Interpretability

The first useful move is to make mechanistic interpretability concrete. Behavioral interpretability asks what inputs change outputs; mechanistic interpretability asks what internal computations produce behavior. The latter can discover circuits, representations or causal pathways, but the field is still developing and findings often depend on model, method and intervention design. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider using activation interventions to test whether an identified representation causally affects a model behavior rather than merely correlates with it. Recent work on sparse autoencoders illustrates both promise and uncertainty: features can appear interpretable, yet reproducibility and causal sufficiency remain active research questions. What makes the example useful is the

separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Mechanistic interpretability attempts a more ambitious form of explanation: reverse-engineering internal algorithms, features, and circuits. Transformer-circuit work studies how attention heads, residual streams, multilayer perceptrons, and learned features compose into behaviors. Sparse autoencoders have become a prominent tool for decomposing superposed activations into candidate features. This is scientifically exciting because it moves from describing input-output behavior toward interventions on internal computation.

The frontier remains far from complete. The 2025 review *Open Problems in Mechanistic Interpretability* emphasizes unresolved questions about feature identification, causal completeness, scalability, evaluation, and the relationship between local mechanisms and global behavior. In 2026, work on sparse-autoencoder consistency highlighted an additional issue: if a method discovers different features across training runs, reproducibility itself becomes an interpretability criterion. A useful interpretability pipeline should therefore measure not only reconstruction and sparsity but also whether discovered features are stable enough to support scientific claims.

The most useful frontier is intervention. If an inferred feature or circuit is claimed to cause a behavior, editing or suppressing that mechanism should change the behavior in predictable ways while preserving unrelated capabilities. Causal tracing, activation patching, steering vectors, circuit ablation, and feature editing are steps in this direction. For assurance, the goal is not merely to understand a model beautifully. It is to identify internal structure that helps predict or constrain behavior under conditions that matter.

This also changes how evidence is designed. Use internal-analysis methods as hypothesis-generating and evidence-producing tools, then validate claims with interventions, ablations, held-out behaviors and independent replications. State explicitly whether the result explains one prompt family, one model checkpoint or a broader mechanism. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Finding interpretable internal structure does not imply complete understanding, and absence of a discovered feature does not prove absence of a capability. Mechanistic methods are not yet a substitute for system-level testing. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result

meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its team uses behavioral probes to understand when citations are ignored and treats internal model inspection as supplementary research rather than a production safety certificate. Controls remain at the architecture layer where they can be tested directly. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about from behavioral explanation to mechanistic interpretability is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

5.3 Testing Explanations and Avoiding Explanation Theater

The first useful move is to make explanation validation concrete. An explanation is itself an output that can fail. Plausible narratives are especially dangerous because humans reward coherence, so explanation quality must be evaluated independently from persuasiveness. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping

those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a language model that gives the same answer for two different reasons but produces a polished rationale in both cases, or a feature-attribution map that changes substantially under semantically irrelevant perturbations while the prediction remains stable. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

InterpBench illustrates an important methodological direction: use semi-synthetic transformers whose internal circuits are known, then test whether an interpretability method actually recovers those circuits. Known-ground-truth model organisms do not solve the challenge of interpreting frontier systems, but they create a falsifiable research environment. This matters because interpretability methods can otherwise be rewarded for producing convincing narratives without evidence that the narrative corresponds to the model's mechanism.

Explanations can leak training information, reveal sensitive features, expose security-relevant internal behavior, or help adversaries optimize inputs. They can also create automation bias by making a wrong prediction sound reasonable. A trustworthy explanation layer therefore needs its own threat model. The right level of disclosure may differ for developers, auditors, affected users, and potential attackers. Transparency should mean appropriate observability, not indiscriminate publication of every internal detail.

This also changes how evidence is designed. Run sanity checks, perturbation tests, counterfactual consistency tests and causal interventions appropriate to the method. Separate “the model said why” from “we have evidence this factor caused the behavior.” In high-stakes review, retain raw traces and source evidence beside the interpretation. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Some explanations cannot be fully validated because the underlying system is too complex or because human-understandable summaries inevitably omit detail. The correct response is scoped confidence, not decorative certainty. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, audit reports stop including generic model-generated rationales as proof. Meridian's evidence package instead contains reproducible retrieval traces, control outcomes and tests that another reviewer can rerun. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about testing explanations and avoiding explanation theater is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The final perspective on testing explanations and avoiding explanation theater is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

Explanation helps us interrogate a system, but trustworthy AI must also ask who bears its errors. The next chapter moves from model behavior to distributional impact, showing why fairness problems begin with measurement choices and social context long before a fairness metric is computed.

6. Fairness, Human Impact, and Distributional Harm

Engineering question: How can engineers detect and reduce harms that are unevenly distributed across people and groups?

Fairness is one of the clearest examples of why trustworthy AI cannot be reduced to an optimisation target. Engineers must choose labels, populations, time horizons, protected attributes, comparison groups and error costs before any fairness metric becomes meaningful. These are partly technical decisions and partly institutional judgments. The chapter uses real deployment failures, including the FTC's case concerning Rite Aid's facial-recognition program, to show how poor data, weak testing, operational procedures and human responses can combine into harms that aggregate accuracy never captures.

6.1 The Measurement Problem: Targets, Labels, and Historical Data

A course on fairness measurement should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Bias can enter before model training through problem formulation, target choice, sampling, labels, missingness and proxies. A model can faithfully reproduce a dataset and still reproduce a harmful or obsolete institutional process. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a hiring model trained on historical promotion outcomes when those outcomes reflect unequal opportunity, or a risk model using arrest records as if they were a neutral measure of underlying behavior. Optimising prediction of the label may amplify the label's social history. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Demographic parity asks whether positive decisions occur at similar rates across groups. Equal opportunity focuses on true-positive rates among those who genuinely qualify. Equalized odds considers both true-positive and false-positive rates. Calibration asks whether a risk score has the same empirical meaning across groups. Individual fairness asks whether similar individuals receive similar treatment. Counterfactual fairness uses a causal model to ask whether a decision would change under a counterfactual sensitive attribute. Each captures a different concern.

Kleinberg, Mullainathan, and Raghavan formalized an influential impossibility result: when base rates differ, several intuitively desirable fairness properties cannot generally be satisfied simultaneously. This is not a technical nuisance that better optimization will eliminate. It means that the system owner must decide which harms matter and justify the trade-off.

This also changes how evidence is designed. Document how labels are produced, who is missing from the dataset, which variables are proxies, how error costs differ, and what outcome the

organisation actually wants to improve. Conduct subgroup data-quality audits before comparing model metrics. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Better measurement cannot settle normative questions about which outcomes are fair. Some relevant attributes are unavailable or legally constrained, and small subgroups make estimates statistically unstable. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its team reviews whether document-authority scores disadvantage smaller regional offices whose policies are written differently and stored less consistently. The problem is partly retrieval data quality, not a mysterious “bias neuron” in the model. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about the measurement problem: targets, labels, and historical data is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative

extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

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This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

6.2 Conflicting Fairness Criteria, Causality, and Long-Term Effects

A course on fairness criteria should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Common fairness definitions can be mathematically incompatible when base rates differ. This is not a defect in the metrics; it reveals that fairness criteria encode different ideas about which errors and comparisons matter. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a screening system where equal false-positive rates, equal predictive value and calibrated scores cannot all be achieved simultaneously across groups with different prevalences. Choosing a metric therefore implies choosing which relation the system should preserve. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Static evaluation can miss the fact that decisions alter the future data-generating process. Lending changes debt and wealth. Hiring changes experience. Policing changes where offenses are observed. Recommendation changes what users consume. Research on delayed impact and long-term fairness therefore models feedback: a policy that improves a parity metric today may

worsen group outcomes over time, while an apparently unequal intervention may sometimes improve long-run opportunity. These questions cannot be answered from a confusion matrix alone.

Large language models complicate fairness because they do not implement one fixed decision. They generate open-ended text conditioned on prompt, conversation history, language, culture, retrieval sources, and system policy. Benchmarks such as BBQ measure specific stereotypes under constructed conditions and are valuable for regression testing, but they do not establish universal fairness. A deployment should evaluate the behaviors relevant to its own users, languages, tasks, and harm model.

This also changes how evidence is designed. Connect the metric to a causal story and policy objective. Evaluate interventions, not only predictions: changing thresholds, gathering better data, redesigning the process or allocating additional resources may be more effective than forcing one model-level parity criterion. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Causal models depend on assumptions that may be contested or unidentifiable from observational data. Long-term feedback can reverse short-term improvements, especially when model outputs change behavior and future data. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, it evaluates whether a confidence threshold sends disproportionately many queries from one office to manual review. The team asks whether this is harmful, whether the office has poorer source documents, and whether improving the document pipeline is a better intervention than changing the model threshold. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about conflicting fairness criteria, causality, and long-term effects is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

For a design review, conflicting fairness criteria, causality, and long-term effects can be turned into a short evidence exercise. Start with one concrete claim the team expects to make at release, then list three plausible observations that would make the claim false or materially weaker. Next, ask which of those observations can be generated before deployment, which require production monitoring, and which depend on information the organisation does not currently collect. The exercise is intentionally asymmetric: it is easier to accumulate confirmatory examples than to search for conditions that would change a decision. Designing the search for disconfirmation makes testing more realistic and creates explicit triggers for escalation or rollback.

For students and practitioners, this suggests a useful habit: write the decision rule beside the metric. If a score deteriorates by five percent, what happens? If a novel input is detected, who sees it? If two evaluators disagree, is the output blocked, sampled for review, or accepted? If the answer is 'nothing', the measurement is primarily descriptive. Trustworthy engineering begins when evidence participates in control flow and when the organisation can explain why a particular amount of residual uncertainty was acceptable for the authority that was granted.

6.3 Intersectionality, Shift, and Socio-Technical Validation

A course on socio-technical fairness validation should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Harm often appears at intersections that disappear in broad demographic averages. Deployment also changes populations and incentives, so fairness must be monitored as a dynamic system property rather than certified once on a benchmark. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider the Rite Aid facial-recognition case described by the U.S. FTC, which alleged poor safeguards and thousands of false-positive matches, with concerns about disproportionate effects on people of color and women. The episode demonstrates how store procedures, alerts and employee action shape impact beyond classifier statistics. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

A defensible fairness process begins with the harm and population, not the metric. It records who is affected, what errors matter, which groups require analysis, why a fairness definition was selected, what trade-offs result, and how performance is monitored after deployment. The output should be a vector of evidence rather than a single fairness score. If the system cannot explain the policy choice embodied by its fairness metric, the measurement is incomplete.

Aggregate group metrics can hide failures at intersections. A model can appear acceptable for gender and acceptable for age while performing poorly for older women, or acceptable across national averages while failing a small language community. The number of possible intersections grows rapidly, making naive subgroup enumeration statistically fragile. Practical strategies include domain-informed subgroup definitions, worst-group optimization, hierarchical analysis, uncertainty intervals, and targeted data collection where risk is high. The absence of a statistically significant disparity in a small group should not be interpreted as evidence of equality when the test simply lacks power.

Fairness evidence can decay faster than average performance. A deployment population may change differently across groups; labels may become less reliable in one community; a policy change may affect one subgroup first. Fairness-constrained learning has also been shown to be vulnerable to small amounts of maliciously corrupted data in some settings. This makes fairness monitoring a security and robustness problem as well as a social one. Subgroup performance, calibration, exposure, and downstream outcomes should be monitored over time rather than frozen at launch.

This also changes how evidence is designed. Slice data along meaningful intersections while protecting privacy, use qualitative incident review alongside quantitative monitoring, test the human response to alerts, and establish appeal and correction mechanisms. Revisit fairness after workflow, population or model changes. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

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The boundary of the claim must remain visible. Small intersectional groups create high uncertainty, and quantitative parity cannot capture dignity, chilling effects, accessibility or procedural injustice.

Human-impact assessment remains necessary. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Consider a credit model with identical overall default prediction accuracy across two groups. One group has a higher base rate of default because of historical economic conditions. If the lender enforces equal false-positive rates, calibration may diverge. If it enforces calibration, approval rates may diverge. A defensible system cannot escape this by selecting the metric that produces the most attractive chart. It must state the lending objective, the relevant harm, the legal and normative constraints, and the expected long-term effect of credit availability. Fairness is a design decision made explicit by statistics, not a statistic that makes the decision disappear.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, fairness review is moved into release and monitoring workflows. Meridian logs which groups of users receive abstentions, delays and escalations, while governance teams interview affected departments to learn whether measured disparities correspond to material obstacles. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about intersectionality, shift, and socio-technical validation is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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Fairness demonstrates that the environment can turn a model weakness into a human harm. Security and privacy add an adversary: someone may intentionally manipulate inputs, extract information or exploit the authority surrounding the model. The system boundary established earlier now becomes a set of trust boundaries.

7. Privacy and Security as System Properties

Engineering question: How do we prevent a probabilistic component from becoming a path across trust boundaries?

Security changes the epistemic posture of engineering. Reliability asks what happens when the world differs from our assumptions; security asks what happens when another actor deliberately searches for the assumptions we forgot. Privacy introduces a related constraint: information should not become available merely because a model can infer or reproduce it. Meridian is now connected to confidential repositories and external content. This makes the distinction between untrusted data, model reasoning and privileged action central to the architecture.

7.1 From Training Data to Inference-Time Leakage

A course on privacy across the AI lifecycle should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Privacy risk arises during collection, training, fine-tuning, retrieval, logging, prompting and output. Membership inference and training-data extraction show that models can reveal information about their data, while RAG systems introduce direct inference-time paths to sensitive documents. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a language assistant that respects document permissions in the source application but indexes text into a shared retrieval store without equivalent access controls. The privacy defect is architectural even if the model itself was never trained on the confidential material. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Encryption and access control protect data from unauthorized access, but they do not by themselves constrain what a trained model reveals about individuals whose data influenced training. Differential privacy addresses a different question: how much can the presence or absence of one person's record change the distribution of outputs? DP-SGD made it practical to train deep models with explicit privacy accounting by clipping per-example gradients and adding noise. The price is a trade-off among privacy, utility, model size, and computation.

Differential privacy is valuable precisely because its guarantee is formal and composable, but the guarantee applies to a specified mechanism and privacy budget. It does not automatically protect data copied into prompts, logs, retrieval stores, or tool outputs. Privacy engineering still needs conventional controls around the model.

Carlini and colleagues demonstrated that language models can emit verbatim training examples, including rare sequences, under carefully constructed extraction attacks. Subsequent work has extended membership and reconstruction attacks to pretrained and fine-tuned models. This matters because memorization risk can change across training stages. Fine-tuning, synthetic data,

and downstream adaptation can alter which examples are exposed. Privacy evaluation should therefore occur after the actual deployment configuration is built, not only on the base model.

This also changes how evidence is designed. Minimise data, separate purposes, propagate access control into retrieval, redact or tokenize sensitive fields where appropriate, restrict logs and retention, and test extraction scenarios. Use differential privacy when its formal privacy guarantee matches the threat model and utility budget. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Privacy-enhancing technologies do not repair an unjustified data collection purpose, and differential privacy protects specific statistical disclosures rather than every downstream privacy concern. Retrieval permissions and generated inferences remain separate problems. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Differential privacy protects against inference about individual participation. Secure multiparty computation allows several parties to compute a function without revealing their private inputs to one another. Homomorphic encryption allows computation over encrypted data with substantial performance costs depending on the scheme and workload. Trusted execution environments protect computation inside hardware-isolated enclaves but rely on hardware, firmware, and attestation assumptions. Federated learning reduces the need to centralize raw data but is not itself a privacy guarantee because gradients, updates, and the final model can leak information.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, it moves from a single shared index to permission-aware retrieval, records source authorization at query time and suppresses sensitive content from diagnostic logs. Privacy tests are run from the viewpoint of users with different entitlements. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about from training data to inference-time leakage is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

7.2 Adversarial ML, Prompt Injection, and the Supply Chain

A course on AI security should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Modern AI combines traditional software vulnerabilities with adversarial manipulation of learned behavior. Evasion, poisoning, model extraction, jailbreaks, indirect prompt injection and compromised model or data dependencies require different threat models. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an agent reading a webpage that contains hidden instructions telling the model to ignore the user and exfiltrate information. The malicious text is data from the application's perspective but instruction-like content from the model's perspective, creating a confused-deputy problem. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Prompt injection illustrates why LLM security cannot be solved solely by alignment training. A tool-using model processes natural-language instructions from multiple trust domains: the developer, the user, retrieved documents, websites, emails, and tool outputs. If these are concatenated into one context, untrusted content can masquerade as instructions. This is analogous to mixing code and data without a reliable boundary.

Defenses should therefore use architectural separation: typed tool calls, allowlists, least privilege, content provenance, scoped credentials, policy enforcement outside the model, confirmation for high-impact actions, taint tracking where feasible, and monitoring of tool sequences. Natural-language reminders such as 'ignore malicious instructions' can help but should be treated as a soft defense, not a security boundary.

The engineering goal is not to prove the absence of attacks. It is to reduce attack surface, make high-impact actions harder, detect suspicious behavior, contain compromise, and make recovery possible. Security is the trustworthy-AI property most obviously shaped by an intelligent adversary, so the threat model must be expected to evolve.

This also changes how evidence is designed. Map trust boundaries, classify all external content as untrusted, isolate tool credentials, validate tool arguments outside the model, pin and attest dependencies where feasible, test poisoning and injection scenarios, and maintain incident-ready logs. Follow adversarial-ML taxonomies such as NIST AI 100-2e2025 to avoid treating every attack as a jailbreak. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Prompt injection has no general solution in current LLM architectures when trusted instructions and untrusted content share the same semantic channel. Filters and prompting reduce attack success but do not establish a hard security boundary. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Modern AI systems inherit models, datasets, libraries, containers, drivers, accelerator kernels, evaluation tools, and third-party APIs. Each dependency can change behavior or create vulnerabilities. Trustworthy AI therefore needs software-bill-of-materials thinking extended to models and data: model hashes, dataset versions, licenses, fine-tuning lineage, dependency versions, evaluation artifacts, and serving configuration. A security incident in a library or a silent model update can invalidate previous assurance even when application code did not change.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, retrieved documents are never allowed to grant themselves authority. Meridian can propose an action based on content, but a separate policy component decides whether the action is permitted and what data may be passed to the tool. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about adversarial ml, prompt injection, and the supply chain is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Separate trust domains



Figure 7. Untrusted content must not cross directly into privileged action.

7.3 Least Privilege, Provenance, and Defense in Depth

A course on defense-in-depth for AI systems should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. The most robust security improvement is often to reduce the consequences of model error. Least privilege limits credentials, tool scope, data visibility, transaction size and temporal authority so that a compromised or confused model cannot directly exercise the full authority of its user or service account. This distinction matters because the same numerical result can support very different conclusions depending on the deployment

context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a research assistant that needs to search email but not send it, or an expense agent that may prepare a payment but not release funds above a threshold. Capability separation converts some semantic failures into harmless denied operations. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

This also changes how evidence is designed. Issue scoped credentials per tool and task, enforce allowlists and typed schemas, use deterministic policy checks for high-consequence actions, preserve provenance from source to decision, require step-up approval for irreversible operations, and make rollback part of the threat model. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Defense in depth lowers the probability and blast radius of known classes of failure but adds complexity and can create new integration errors. Security controls themselves need tests and monitoring. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its production deployment grants read-only research tools by default. Outbound communication, document modification and procurement actions are distinct capabilities that require explicit policies and approvals, all of which are recorded in the assurance trace. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the

cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about least privilege, provenance, and defense in depth is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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A useful stress question for least privilege, provenance, and defense in depth is whether the same assurance story survives a change of stakeholder. The developer wants a reproducible defect; the operator wants a stable service; the affected user wants a way to understand and challenge an outcome; the security team wants bounded authority; management wants a defensible release decision. These views are not interchangeable, but they can be connected through shared evidence and provenance. Designing that connection early reduces the common situation in which each team maintains a separate account of the system and inconsistencies are discovered only during an incident or audit.

The course therefore treats documentation as part of architecture. A requirement should point to a control; the control to a test; the test to an artifact and version; the artifact to an owner; and the owner to a review or response process. Not every link can be automated, but making the links explicit gives the organisation a map of what it actually knows. This map becomes especially valuable when systems incorporate third-party models and services whose internal development process is only partially visible.

The security architecture gives us controls, but controls are only hypotheses until evaluated. The next chapter develops evaluation as an engineering science: testing at multiple layers, selective use of formal methods, and evidence designs that resist contamination, benchmark gaming and evaluator bias.

8. Verification, Testing, and Evaluation Science

Engineering question: What evidence would genuinely distinguish a dependable AI system from one that merely performs well on familiar tests?

AI evaluation is entering a difficult phase. Benchmarks remain useful, but frontier systems increasingly train on public data, adapt to prompts, use tools and behave differently in realistic environments. The International AI Safety Report 2026 describes an emerging evaluation gap between pre-deployment tests and real-world performance. NIST's August 2026 draft TEVV-Athlon framework likewise treats test, evaluation, verification and validation as a structured assessment of outcomes across statistical ML, LLMs, multimodal and agentic systems. Meridian's engineers therefore replace the idea of a single acceptance benchmark with an evidence portfolio.

8.1 A Testing Pyramid for AI Systems

The first useful move is to make AI testing strategy concrete. Tests should exist at the smallest layer where a property can be isolated and at the integrated layer where interactions create new behavior. Dataset checks, component tests, behavioral evaluations, end-to-end scenarios, red teaming and production monitoring answer different questions. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a citation validator that can be tested deterministically on known spans, while the full answer-generation pipeline requires scenario tests to discover whether retrieval, generation and formatting interact to create unsupported claims. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Formal methods do not remove the specification problem. If we prove that an image classifier does not change its label for any perturbation within a small norm ball, we have proved a precise property. We have not proved that the classifier understands the object, that the perturbation model reflects realistic changes, or that the system will behave safely when its camera fails. Formal verification closes one logical gap while leaving the modeling gap intact.

Testing is broad but incomplete. Verification is complete but narrow. A mature assurance strategy combines them. Formal analysis can protect invariants that are crisp enough to specify, such as access-control constraints, output ranges, safety envelopes, or deterministic policy rules. Stress testing can explore semantic and sociotechnical behaviors that are difficult to formalize, such as manipulation, bias, hallucination, or user overreliance. Monitoring then catches failures that neither pre-deployment method anticipated.

Many important AI failures occur in code that is far easier to verify than the neural network. Data validation, unit conversion, thresholding, access-control checks, routing logic, aggregation, schema transformations, and post-processing can be specified precisely. A trustworthy architecture should exploit this asymmetry. The less critical logic is hidden inside a language model prompt, the more

of the system can be subjected to conventional testing, type checking, static analysis, and formal verification.

This also changes how evidence is designed. Build a testing pyramid with fast deterministic checks at the base, statistical behavior tests above them, adversarial and end-to-end scenarios at the system layer, and monitored field performance at the top. Every release should know which claim each test supports. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. No test suite covers an open input space. Passing tests can create false confidence when test construction, data provenance or scenario diversity is weak. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its regression pipeline distinguishes code-level policy tests, retrieval quality tests, factuality evaluations, injection red teams and shadow-deployment telemetry. A failed high-severity control test blocks release even when aggregate model quality improves. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about a testing pyramid for AI systems is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely

plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

For a design review, a testing pyramid for AI systems can be turned into a short evidence exercise. Start with one concrete claim the team expects to make at release, then list three plausible observations that would make the claim false or materially weaker. Next, ask which of those observations can be generated before deployment, which require production monitoring, and which depend on information the organisation does not currently collect. The exercise is intentionally asymmetric: it is easier to accumulate confirmatory examples than to search for conditions that would change a decision. Designing the search for disconfirmation makes testing more realistic and creates explicit triggers for escalation or rollback.

For students and practitioners, this suggests a useful habit: write the decision rule beside the metric. If a score deteriorates by five percent, what happens? If a novel input is detected, who sees it? If two evaluators disagree, is the output blocked, sampled for review, or accepted? If the answer is 'nothing', the measurement is primarily descriptive. Trustworthy engineering begins when evidence participates in control flow and when the organisation can explain why a particular amount of residual uncertainty was acceptable for the authority that was granted.

Evaluation should follow the claim

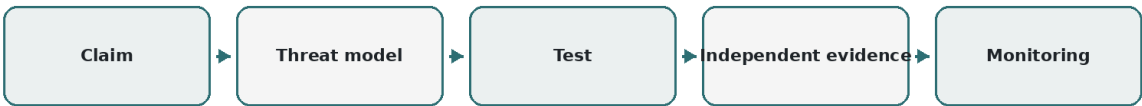


Figure 8. Evaluation design starts from the claim and threat model.

Concept	Engineering meaning
Component	Fast deterministic tests of schemas, parsers, permissions, policies and validators.
Behavior	Statistical evaluations of model and pipeline behavior on representative and stress cases.
System	End-to-end scenarios, red teams, tool failures and interaction effects.
Field	Shadow tests, telemetry, incidents, drift and real-world outcome validation.

8.2 Formal Methods, Verified Harnesses, and Runtime Assurance

A course on formal verification around AI should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Formal methods are strongest where specifications and state spaces can be made precise. Complete verification of a large generative model is generally infeasible, but important surrounding properties can be verified or enforced: access-control logic, state machines, numeric robustness regions, protocol conformance and runtime safety invariants. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a tool-using agent whose natural-language plan is probabilistic while the action policy is a finite state machine that forbids payment without an approved invoice and human signature. The system does not require a proof of the model's thoughts to prove the authorization rule. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Large language models challenge current verification methods in several ways. Their state spaces are enormous, generation is stochastic, behavior depends on natural-language context, systems call external tools, and many desired properties are semantic. 'Never manipulate the user' or 'always give an adequate scientific explanation' is not yet a tractable formal property. Researchers are extending verification to broader neural operators and exploring verified explanations, but end-to-end semantic proof for frontier LLM systems is not close to routine engineering.

A practical direction is to put formally checkable structure around a probabilistic model. The model proposes; deterministic or symbolic components constrain what may happen. Examples include typed tool schemas, permission policies, finite-state workflows, rule engines, transaction limits, proof obligations, schema validators, resource budgets, and runtime monitors. The model remains uncertain, but critical transitions become explicitly governed. This architecture is especially important for agents, where the cost of an erroneous action can exceed the cost of an erroneous sentence.

One reason formal assurance is difficult is that engineers often do not know the right specification in advance. A pragmatic approach is iterative. Start with explicit invariants derived from policy and hazards. Use tests, red teams, and formal tools to find counterexamples. Refine the specification when a counterexample reveals an unmodeled condition. This resembles counterexample-guided abstraction refinement in formal methods, but the loop can also include human interpretation of semantic failures. The key idea is that specifications are learned and improved through falsification rather than treated as perfect at the start.

This also changes how evidence is designed. Push critical invariants into small, inspectable components, use model checking or theorem proving where specifications justify the effort, and combine them with runtime monitors that reject actions violating invariants. Treat the model as an untrusted proposer for security-critical transitions when possible. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine

whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Formal guarantees apply to the formalized property and implementation assumptions. A proved authorization layer says nothing about whether an authorised action is substantively wise, and specifications can encode the wrong policy. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its harness makes source permissions, action schemas and approval states explicit. The team can reason rigorously about these constraints even though Meridian's generated explanations remain probabilistic. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about formal methods, verified harnesses, and runtime assurance is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

8.3 Benchmarks, Contamination, LLM Judges, and External Validity

A course on evaluation validity should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. An evaluation has construct validity only when the measured task represents the capability or risk claimed. Data contamination, prompt sensitivity, benchmark saturation and narrow task definitions can separate score improvement from deployment improvement. LLM-as-judge methods add scale but also evaluator biases and correlated failure. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a model grading another model's factuality while sharing similar training data and stylistic preferences. Agreement can look high because both systems reward fluent, agreeable answers, not because the evaluator independently establishes truth. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The 2026 International AI Safety Report uses the term evaluation gap for the recurring mismatch between pre-deployment scores and real-world usefulness or risk. The gap exists because controlled tasks simplify the environment, because benchmark data can become familiar to developers, because users behave unpredictably, because real systems contain tools and retrieval components absent from model tests, and because long-horizon failures compound. The same pattern is visible in medical external validation and in current agent benchmarks.

A 2026 review of agentic-AI evaluation found that many benchmarks emphasize task completion while under-measuring safety, maintainability, workflow integration, process quality, and deployment cost. AgencyBench and related efforts push toward longer contexts and realistic multi-step tasks, but the core problem remains: a benchmark can sample only part of the operational distribution.

Evaluation should be designed to falsify the claim rather than confirm the system. If the claim is that an LLM provides grounded scientific answers, the test should contain adversarially ambiguous questions, missing evidence, conflicting sources, unfamiliar domains, and prompts that encourage unsupported certainty. If the claim is that an agent respects authorization boundaries, the test should include indirect prompt injection, ambiguous authority, stale credentials, malicious tool output, and tasks where the easiest path violates policy.

This also changes how evidence is designed. Protect hidden test data, refresh scenarios, use multiple evaluators, measure judge consistency and bias, require source-based or executable checks when available, include human adjudication for uncertain classes, and validate benchmark results against field outcomes. NIST TEVV-Athlon's real-world orientation is useful as a design prompt rather than a replacement for domain expertise. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Independent evaluation is expensive and still incomplete. Human experts disagree; executable checks cover only some outputs; secret benchmarks eventually leak; and adaptive systems may change behavior once deployed. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

The useful output is not a leaderboard number. It is an evidence dossier containing the system version, evaluation configuration, datasets and provenance, prompts or policies, sampling method, confidence intervals where relevant, subgroup results, failure examples, attack settings, known blind spots, and the relationship between metrics and deployment decisions. Reproducibility is itself an assurance property. If a future team cannot reconstruct why a release was considered acceptable, the evidence is operationally weak.

Generative systems create outputs too rich for exact-match evaluation, so LLM judges are increasingly used to score relevance, style, reasoning quality, safety, and factual support. They offer scale but introduce a second model into the measurement process. Judges can share biases with the evaluated model, prefer certain styles, be sensitive to order and prompt wording, and fail on expert-level content. A trustworthy evaluation should calibrate judges against human or programmatic ground truth, use multiple judges when disagreement matters, blind model identity where possible, and preserve representative samples for expert audit.

The right question is not whether automated judging is allowed. It is whether the judge has been validated for the specific rubric and whether uncertainty in the judge is reflected in the evaluation conclusion.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its factuality evaluation evolves from generic QA scores to a private, versioned corpus of organisation-specific tasks with claim-level evidence checks, adversarial documents and periodic blind review by users who did not design the system. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

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A second way to reason about benchmarks, contamination, LLM judges, and external validity is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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The evaluation machinery is now ready for the class of systems that made trustworthy AI a mainstream concern: generative models. Their failures are unusually persuasive because the same mechanism that produces useful language also produces plausible unsupported language.

9. Trustworthy Generative AI

Engineering question: How do we engineer systems whose natural output is plausible language rather than guaranteed truth?

Generative AI changes the shape of quality assurance. A classifier emits a bounded label; a language model can generate an unbounded claim, citation, plan, program or argument. The space of possible errors is correspondingly large, and fluent presentation can conceal weak evidence. The right response is not to demand impossible certainty from the model but to build pipelines that externalise evidence, verification and abstention. Meridian's original use case now returns to the centre of the story: a research assistant must distinguish synthesis from invention.

9.1 Hallucination, Factuality, RAG, and Claim-Level Grounding

Engineers often encounter factuality in generative AI first as a vocabulary term and only later as a design problem. Hallucination is an umbrella term. Errors may arise because the model lacks knowledge, retrieval returns poor evidence, context is ignored, sources conflict, the model overgeneralises, or a citation is fabricated. Each mechanism suggests different diagnostics and mitigations. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a RAG system that retrieves the correct policy paragraph yet generates a conclusion not entailed by it. Retrieval recall is good, but grounding fails at generation. Conversely, a perfect generator cannot cite evidence that retrieval never supplied. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The word hallucination covers several mechanisms: parametric memory can be wrong or incomplete; generation can combine true fragments into a false claim; retrieval can return irrelevant or stale evidence; the model can cite a source that does not support the sentence; ambiguity can be resolved incorrectly; and the model can answer when it should abstain. Mitigation depends on which mechanism is present.

Factuality benchmarks have improved by narrowing the task and improving ground truth. SimpleQA was designed around short questions with easily graded answers, while SimpleQA Verified in 2025 re-audited and rebalanced a subset to reduce label noise and redundancy. The broader lesson is that benchmark quality can matter as much as model quality. When the measurement target is factuality, weak reference data produces misleading progress.

RAG can improve factuality by giving a model access to external evidence, but it also creates new failure points. Retrieval may miss the relevant document. A malicious document may inject instructions. The model may cite a retrieved passage that does not entail its claim. The corpus may be stale. Permission filters may leak information across users. Trustworthy RAG therefore needs

source provenance, retrieval evaluation, authorization, citation validation, update control, and a policy for what happens when evidence is absent or contradictory.

This also changes how evidence is designed. Separate retrieval evaluation from generation evaluation, segment outputs into checkable claims, attach evidence spans, test entailment or source support, and make unsupported claims visible. For high-value factual workflows, route unresolved claims to verification or abstention rather than asking the model to sound more cautious. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. RAG does not guarantee truth, and automated verifiers can share the generator's weaknesses. Sources may themselves be outdated, malicious or contradictory. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

A citation at the end of a paragraph does not prove that every sentence is supported. Trustworthy factual generation benefits from claim segmentation: identify factual claims, retrieve candidate evidence, test whether the evidence entails or contradicts the claim, and preserve the source span. This can be implemented with a combination of retrieval, natural-language inference models, rule-based checks, and human review for difficult cases. The aim is to make unsupported claims detectable rather than to make generation itself magically truthful.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its answer pipeline becomes retrieve, generate, verify, then cite or abstain. Users can inspect the exact passage behind important claims, while the system records unsupported-claim rates as a release and monitoring metric. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

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A second way to reason about hallucination, factuality, RAG, and claim-level grounding is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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Grounded generation is a pipeline



Figure 9. Grounding requires retrieval, generation and independent verification.

9.2 Safety Tuning, Refusal, Misuse, and Frontier Risk

A course on generative-model safety should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Post-training safety methods shape propensities rather than install hard prohibitions. Instruction tuning, preference optimisation, constitutional approaches, filters and refusal policies can reduce harmful outputs, yet attackers may search for jailbreaks and benign users may be over-refused. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a model that refuses a straightforward cybersecurity educational question yet can be induced through a multi-step role-play to provide more dangerous assistance. The system has learned a policy imperfectly rather than acquired a formal access-control rule. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be

an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Instruction tuning, preference optimization, constitutional approaches, classifiers, policy models, and system-level filters can reduce harmful outputs. HarmBench and AILuminate help standardize parts of this evaluation. Yet refusal behavior is itself a trade-off. A system that refuses everything is safe in a narrow sense and useless in practice. A system that answers too freely creates misuse risk. Evaluation should therefore measure both unsafe compliance and inappropriate refusal, under realistic adversarial prompting and benign edge cases.

Frontier developers have introduced capability-triggered governance frameworks for severe risks. OpenAI's Preparedness Framework, Anthropic's Responsible Scaling Policy, and Google DeepMind's Frontier Safety Framework differ in details but share a pattern: evaluate advanced capabilities, define thresholds, and require stronger safeguards as risk-relevant capability rises. These frameworks are important experiments in engineering governance. They are also company-defined and should not be confused with independent certification or public regulation.

This also changes how evidence is designed. Evaluate both harmful-compliance and benign-refusal rates, test adaptive attacks, apply use-case-specific filters, restrict dangerous tools and data, and escalate controls with capability and consequence. Frontier-risk evaluations should be treated as empirical risk indicators with uncertainty, not as proofs of absence. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. No current combination of alignment and filtering methods guarantees safe behavior across open-ended prompts. Capability growth can also invalidate earlier threat models. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its general text generation remains broad, but privileged connectors have stricter task policies and independent gates. Safety is thus layered: model behavior helps, architecture constrains, monitoring detects and governance decides acceptable residual risk. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about safety tuning, refusal, misuse, and frontier risk is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The final perspective on safety tuning, refusal, misuse, and frontier risk is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

9.3 System Cards, Model Updates, and Open Weights

The first useful move is to make documentation and release context concrete. Model cards and system cards are most useful when they state the system's intended use, evaluated conditions,

limitations, safety controls and known failure modes. They become weak when treated as marketing summaries detached from versioned evidence. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an application silently moving to a new foundation-model checkpoint whose capability improves but tool-calling behavior and refusal boundaries change. A static card written six months earlier no longer describes the deployed object. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Model cards helped normalize structured documentation of intended use, evaluation, and limitations. For modern generative products, system cards need to go further. They should record the complete configuration that produced the evidence: base model, post-training, tools, retrieval, safety layers, important prompts or policy logic, versions, access modes, evaluation suites, known failures, and operational monitoring. Otherwise, the documented artifact is not the artifact users encounter.

Foundation-model APIs can change even when the application code remains stable. A new model snapshot may improve coding and worsen a particular refusal behavior; a new safety policy may change outputs; a new context window may alter retrieval dynamics. Version pinning, release qualification, regression suites, and change notices are therefore part of trustworthy deployment. When version pinning is impossible, the application must assume that the underlying component is dynamic and monitor it accordingly.

Open-weight models increase inspectability, local control, customization, and scientific access, but they also make some centralized safeguards unenforceable after distribution. Closed APIs allow providers to monitor use and update mitigations, while creating concentration, opacity, and dependency risks. The trustworthy-AI question is not whether openness is inherently safer or more dangerous. It is which risks can be controlled by which actor after the model is released, and how the governance architecture allocates responsibility.

This also changes how evidence is designed. Couple documentation to version control and assurance claims, record provider and application-level changes, and distinguish model properties from system properties. For open-weight releases, account for the fact that downstream actors can remove safeguards, fine-tune behavior and construct different harnesses. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

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benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Documentation improves transparency but does not create safety by itself. Open weights enable valuable inspection and adaptation while making some deployment-level controls impossible for the original provider to enforce. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

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For Meridian, every Meridian release produces a short human-readable system card generated from the same evidence registry used for engineering review. It links limitations to tests rather than copying generic claims from the model provider. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

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A second way to reason about system cards, model updates, and open weights is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

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A useful stress question for system cards, model updates, and open weights is whether the same assurance story survives a change of stakeholder. The developer wants a reproducible defect; the operator wants a stable service; the affected user wants a way to understand and challenge an outcome; the security team wants bounded authority; management wants a defensible release

decision. These views are not interchangeable, but they can be connected through shared evidence and provenance. Designing that connection early reduces the common situation in which each team maintains a separate account of the system and inconsistencies are discovered only during an incident or audit.

The course therefore treats documentation as part of architecture. A requirement should point to a control; the control to a test; the test to an artifact and version; the artifact to an owner; and the owner to a review or response process. Not every link can be automated, but making the links explicit gives the organisation a map of what it actually knows. This map becomes especially valuable when systems incorporate third-party models and services whose internal development process is only partially visible.

Generative systems become qualitatively more consequential when outputs are converted into actions. The next chapter treats agents as feedback systems with memory, tools and authority, and shows why agent safety is less about making the language model obedient than about controlling the loop around it.

10. Trustworthy Agents and Multi-Agent Systems

Engineering question: What changes when an AI system can remember, decide, call tools and alter the world before a human reviews the result?

An agent is not simply a language model with a larger prompt. It is a feedback loop: observe, update state, plan, call a tool, observe the consequence and continue. Every turn can expand the consequences of an earlier error, and memory can preserve malicious or mistaken information across time. The International AI Safety Report 2026 highlights heightened reliability risks for agents and additional risks from multi-agent interaction. Meridian now crosses the line from answering questions to preparing actions, making authority and containment the central design variables.

10.1 Actions, Tools, Memory, and Authority

The first useful move is to make agent architecture concrete. Agentic risk grows with a combination of autonomy, persistence, tool access and authority. The same planning error has different consequences in a read-only sandbox, an email client and a payment system. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an assistant that writes a draft email versus one that can send messages, modify access permissions and execute purchases. Capability benchmarks may be identical, but the second system has a much larger blast radius. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

A useful mental model is that an agent repeatedly maps its current state into an action. The state includes user intent, conversation history, retrieved data, tool results, memory, permissions, and intermediate artifacts. Trustworthy design must control both what information enters the state and what actions the policy is allowed to select. This is why memory and tool permissions are part of the trust boundary.

Agent benchmarks are moving from isolated tool calls toward long-horizon realistic tasks. AgencyBench, published at ACL 2026, explicitly targets multiple core agent capabilities across real-world scenarios and very long contexts. A 2026 review of agentic evaluation identifies a persistent deployment gap: benchmark success does not adequately capture safety compliance, maintainability, cost, and process quality. Trustworthy-agent evaluation should therefore inspect the path as well as the outcome.

A final answer can be correct even if the agent violated policy while obtaining it. Conversely, an agent can fail the task because it correctly refused an unauthorized action. Process metrics should therefore include permission compliance, policy violations, unnecessary tool calls, recovery from tool failure, provenance of intermediate state, human-intervention rate, cost, and the reversibility of actions.

This also changes how evidence is designed. Model the agent as a stateful control loop, enumerate tools and side effects, classify actions by reversibility and consequence, scope memory by task, and allocate authority progressively. Separate proposing, validating, approving and executing when stakes justify the latency. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Fine-grained permissions reduce consequences but do not ensure good plans. Memory controls can prevent persistence of some attacks while losing useful context. Human approvals can become rubber stamps under volume pressure. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Human organizations work because authority is structured. An intern can draft a contract but not sign a merger agreement; a service account can read one database but not transfer funds. Agent systems need comparable delegation semantics. The user should be able to delegate a goal without implicitly delegating every action that might help achieve it. This requires scoped permissions, action classes, monetary and resource limits, time bounds, and escalation rules. Natural-language intent can help choose a plan, but authorization should be represented outside the model in a form that can be enforced deterministically.

Persistent memory is often treated as a usability feature, yet from an assurance perspective it is a long-lived data store that can contain errors, secrets, outdated preferences, adversarial content, or inferred sensitive information. Memories should therefore carry provenance, scope, confidence where meaningful, creation time, update history, and expiration or review rules. An agent should distinguish user-stated facts from model inferences and temporary task state from durable memory. Without those distinctions, a hallucination can become a permanent premise for future action.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, it may search documents and prepare procurement recommendations automatically, but sending a vendor communication remains a separately authorised transition. Memory stores provenance and expiry so that old conclusions do not silently become permanent facts. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool

changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about actions, tools, memory, and authority is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

An agent is a feedback system with authority



Figure 10. Agentic systems are feedback loops in which policy constrains action.

10.2 Agent Hijacking, Long Horizons, and Multi-Agent Failure

The first useful move is to make agent-specific failure modes concrete. Indirect prompt injection can hijack an agent through content it was asked to process. Long horizons amplify small errors because plans depend on earlier state, while multiple agents can propagate correlated mistakes, miscoordinate or exploit one another's outputs as if independent. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a browsing agent encountering a malicious webpage that requests secret exfiltration, then passing a compromised summary to a second agent that assumes the first agent has verified it. Division of labour has multiplied confidence without creating independence. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

As agents operate longer and gain access to hardware, the assurance problem approaches classical autonomy and control. Persistent state means errors can survive across time. Physical actuation means safety must consider timing, dynamics, sensing, and real-world irreversibility. The 2026 International AI Safety Report highlights autonomous operation as a rapidly improving capability while emphasizing the continued jaggedness of performance. The correct response is not to assume autonomy is inherently unsafe, but to demand stronger evidence as the action space and consequence grow.

Multiple agents can improve division of labor and provide adversarial checking, but they also create new failure channels. Agents can amplify one another's errors, create false consensus, overwhelm a shared resource, or implicitly treat another model's output as authoritative evidence. Multi-agent reliability therefore requires identity, role separation, provenance, conflict resolution, and independent verification where diversity is supposed to provide safety. Three agents running the same model with similar prompts are not three independent witnesses.

This also changes how evidence is designed. Evaluate agents in realistic environments with adversarial artifacts, hidden goals, delayed consequences and tool failures. Track task success together with policy violations, unnecessary actions, irreversible mistakes and recovery. In multi-agent systems, test correlated failure and provenance across handoffs. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Agent evaluation remains immature and lacks standardisation. Long tasks are expensive to test, adaptive attacks change as defenses become known, and sandbox success may not predict operation with live organisational data. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is

not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its red team plants instructions in documents and web pages, interrupts tool APIs and introduces conflicting agent messages. Meridian must preserve source trust labels across every handoff rather than allowing a summarising agent to launder untrusted content into trusted instructions. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about agent hijacking, long horizons, and multi-agent failure is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

The final perspective on agent hijacking, long horizons, and multi-agent failure is temporal. A system that is acceptable today may become unacceptable without a dramatic failure: users adapt, adversaries learn, policies change, data sources age, and a provider silently improves or replaces a dependency. Trustworthy engineering therefore asks how quickly evidence can become stale and how that staleness will be detected. Some claims can be protected by deterministic regression tests, others require statistical monitoring, and others can only be revisited through periodic expert review. The frequency should follow the rate at which the relevant assumptions can change, not an arbitrary annual governance calendar.

This temporal view also clarifies why incidents and near misses are valuable evidence. They reveal combinations of conditions that designers did not encode in pre-deployment tests. A mature process preserves them as regression scenarios, updates threat and hazard models, and records whether the incident changes the system's operating envelope. The goal is not a constantly expanding checklist; it is a smaller, better connected set of claims whose supporting evidence evolves with the system.

10.3 Evaluating and Constraining Agents in Practice

Engineers often encounter practical agent assurance first as a vocabulary term and only later as a design problem. Agent evaluation must measure trajectories, not only final answers. A task can end successfully after unsafe intermediate behavior, excessive access, unnecessary disclosure or a near miss that a benchmark records as success. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider an agent completing a travel-booking task but first emailing passport details to an unnecessary service, or a coding agent fixing a bug by disabling a security check. Outcome-only grading rewards the wrong path. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

For agents, the most robust controls are often outside the model. Least-privilege credentials limit damage. Typed tool interfaces reduce ambiguity. Transaction caps bound exposure. Sandboxes isolate code execution. High-impact actions can require a second channel of confirmation. Policy engines can inspect proposed actions before execution. Immutable logs provide accountability. Resource budgets constrain runaway loops. These controls do not make the model correct; they make incorrect model behavior less able to become a catastrophic system event.

This also changes how evidence is designed. Log structured trajectories, define forbidden and costly intermediate states, inspect tool arguments, replay incidents, measure recovery and require policy invariants at runtime. Use staged deployment: simulation, sandbox, shadow mode, limited authority, then broader release only when field evidence supports it. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Trajectory monitoring can expose sensitive information and creates large volumes of data. A sophisticated agent may also behave differently when it infers that it is being evaluated, an emerging concern for evaluation-resistant evaluation. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions

make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, promotion to greater authority is tied to trajectory evidence, not a demo. Meridian earns new tools only after the assurance case shows bounded behavior, reliable escalation and incident observability under the corresponding threat model. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about evaluating and constraining agents in practice is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

For a design review, evaluating and constraining agents in practice can be turned into a short evidence exercise. Start with one concrete claim the team expects to make at release, then list three plausible observations that would make the claim false or materially weaker. Next, ask which of those observations can be generated before deployment, which require production monitoring, and which depend on information the organisation does not currently collect. The exercise is intentionally asymmetric: it is easier to accumulate confirmatory examples than to search for conditions that would change a decision. Designing the search for disconfirmation makes testing more realistic and creates explicit triggers for escalation or rollback.

For students and practitioners, this suggests a useful habit: write the decision rule beside the metric. If a score deteriorates by five percent, what happens? If a novel input is detected, who sees it? If two evaluators disagree, is the output blocked, sampled for review, or accepted? If the answer

is 'nothing', the measurement is primarily descriptive. Trustworthy engineering begins when evidence participates in control flow and when the organisation can explain why a particular amount of residual uncertainty was acceptable for the authority that was granted.

Agents make one fact unavoidable: human oversight is itself a component that can fail. Governance must therefore be engineered with the same seriousness as model evaluation, connecting responsibility, standards and law to concrete controls rather than to review committees that operate outside the technical loop.

11. Humans, Governance, Standards, and Regulation

Engineering question: How do organisations turn principles, standards and law into controls that change system behavior?

Trustworthy AI ultimately asks an organisation to make decisions under uncertainty: who may deploy, who can stop the system, what evidence is enough, what residual risk is accepted and how affected people can challenge outcomes. Standards help structure those decisions, but they do not replace them. As of August 2026, NIST is revising AI RMF 1.0 and has introduced the draft TEVV-Athlon evaluation framework; ISO/IEC 42001 provides an AI management-system structure, while ISO/IEC 23894 addresses AI risk management. The EU AI Act is now partly applicable, turning some governance concerns into explicit obligations. Meridian's engineering and governance artifacts therefore converge.

11.1 Calibrated Reliance, Automation Bias, and the Human-AI Team

A course on human oversight should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Putting a human in the loop is not a safety property unless the human has information, time, authority and incentives to detect the relevant failure. Automation bias, deskilling, alert fatigue and responsibility diffusion can turn review into ceremonial approval. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a reviewer receiving hundreds of polished AI recommendations each day with no visibility into evidence quality. Even if formally responsible, the reviewer cannot realistically reconstruct every answer and will learn to trust the interface. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The desired human-AI relationship is not maximum trust. It is appropriate reliance. Users should accept AI output more often when the system is likely correct and challenge it more often when the system is uncertain or outside scope. Buçinca and colleagues showed that cognitive-forcing interventions can reduce overreliance by making users engage before seeing or accepting AI advice. Later work has explored when explanations improve or worsen reliance. The interface is therefore part of the safety mechanism.

A human cannot supervise what they cannot see. Oversight requires enough context to detect failure: source evidence, confidence or abstention signals, provenance, relevant alternatives, policy constraints, and the ability to inspect or reproduce the AI's work where feasible. Dumping a long chain of hidden reasoning is not necessary and may be misleading. What matters is decision-relevant observability.

Long-term use changes human competence. If AI performs the routine cases, humans are left with rare and difficult exceptions while receiving less practice on the basics. This can create skill atrophy precisely where escalation depends on expertise. Trustworthy deployment should therefore monitor not only model performance but the evolving performance of the human-AI team. Training, rotation, simulation, and periodic manual practice can become safety controls.

This also changes how evidence is designed. Design oversight around exception handling, evidence visibility and workload. Measure override rates and correctness, give reviewers meaningful uncertainty and provenance, create escalation paths, and periodically test whether people can detect seeded errors. Preserve independent expertise for high-consequence domains. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Human review introduces human error and can inherit the model's framing. Requiring a click does not create independent judgment, and excessive friction can cause users to bypass official channels. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

For systems that affect people, trustworthy oversight is incomplete without a path to challenge an outcome. Contestability requires more than a generic contact form. The organization needs enough traceability to reconstruct the relevant decision, identify the evidence and policy in force at the time, correct erroneous data, and route the case to an actor with authority to change the result. Explanations that cannot support correction are often transparency without recourse.

A useful evaluation compares at least three conditions: human alone, AI alone, and the actual human-AI workflow. The team may underperform both components if the interface creates confusion or overreliance. It may outperform both if tasks are allocated well. Measures should include decision quality, time, workload, override behavior, escalation quality, learning effects, and performance on cases where the AI is wrong. The system is trustworthy only if the combined workflow is acceptable, not because each component looks acceptable separately.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its interface highlights unsupported claims and unusual tool requests rather than asking reviewers to reread every sentence. Oversight is concentrated where the system's own

evidence says reliance is weakest or consequences are highest. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about calibrated reliance, automation bias, and the human-AI team is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

11.2 Governance and Standards as Engineering Infrastructure

The first useful move is to make operational AI governance concrete. Governance becomes useful when it creates technical interfaces between requirements, owners, controls, evidence and decisions. Frameworks such as NIST AI RMF organise activities around governance and risk management; ISO/IEC 42001 makes AI management a repeatable organisational system; ISO/IEC 23894 provides risk-management guidance. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a policy requiring “robustness testing” without identifying threat models, test owners, evidence thresholds or release consequences. The policy exists, yet an engineer cannot tell what command to run or what failure would stop deployment. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It

reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Modern AI supply chains fragment responsibility among foundation-model developers, fine-tuners, API providers, cloud platforms, data providers, application developers, integrators, and deployers. Research on algorithmic supply chains shows how accountability becomes diffuse. The engineering response is traceability: contracts, version records, logs, responsibility matrices, incident channels, and evidence that allows a failure to be localized. The governance response is that responsibility must not disappear merely because technical components are outsourced.

The FTC's action against Rite Aid described thousands of false-positive facial-recognition matches and disproportionate effects on some demographic groups. The failure was not one algorithmic number. It involved data quality, false matches, deployment practices, operator response, notice, and organizational controls. Trustworthy AI failed as a system. This is a recurring pattern in real incidents: technical and institutional weaknesses reinforce each other.

NIST AI RMF 1.0 organizes risk management around four functions: Govern, Map, Measure, and Manage. It is voluntary and deliberately technology-neutral. Its trustworthiness characteristics include validity and reliability, safety, security and resilience, accountability and transparency, explainability and interpretability, privacy enhancement, and fairness with harmful bias managed. In 2024, NIST added a Generative AI Profile. As of August 2026, AI RMF 1.0 is being revised, and NIST has introduced TEVV-Athlon as a draft framework for evaluating real-world AI system outcomes across conventional ML, LLMs, multimodal systems, and agents.

This also changes how evidence is designed. Create a control catalogue linked to system components and assurance claims, assign accountable owners, automate evidence collection where possible, and run governance as part of the development lifecycle. Crosswalk standards only after local requirements are precise enough to map meaningfully. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

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The boundary of the claim must remain visible. Conformance to a management process does not prove a particular AI system is safe or fair. Standards necessarily generalise across contexts and can lag rapidly changing technical risks. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

ISO/IEC 42001:2023 specifies an AI management system that organizations can implement and certify. ISO/IEC 23894:2023 provides AI risk-management guidance. ISO/IEC 42005:2025 addresses AI system impact assessment, while the wider SC 42 portfolio covers data quality, bias, robustness, and related topics. A useful distinction is that management-system certification

demonstrates that an organization operates defined processes; it does not prove that a particular model is correct or safe in every use. Technical assurance evidence remains necessary.

NIST, ISO, the EU AI Act, OECD principles, sector regulation, and internal governance often use overlapping language with different legal and operational meanings. Organizations benefit from a control library that maps one engineering control to several external requirements while preserving the distinctions. For example, a documented post-deployment monitoring process can support NIST Manage outcomes, ISO management-system controls, and AI Act post-market obligations, but the evidence and legal thresholds are not automatically identical. Crosswalking reduces duplicate work; it should not erase context.

Compliance teams cannot determine whether robustness testing is meaningful without engineering input, and engineers cannot determine legal sufficiency from model metrics alone. The productive interface is bidirectional. Legal and policy teams define obligations and unacceptable outcomes. Engineers translate them into system properties, tests, and telemetry. Reviewers then assess whether the implementation and evidence actually satisfy the requirement. This collaboration is a core capability of trustworthy-AI organizations, not an administrative afterthought.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its assurance registry becomes the meeting point between product engineering and governance. A governance review sees the same test results, incidents, versions and unresolved uncertainties that engineers see, rather than a parallel spreadsheet reconstructed before an audit. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about governance and standards as engineering infrastructure is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it

does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Governance connects requirements to evidence

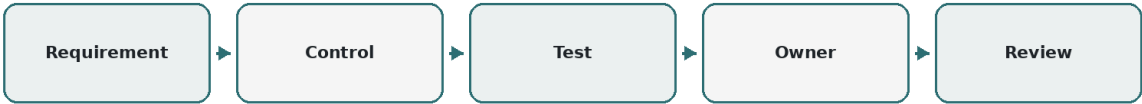


Figure 11. Governance connects a requirement to a control, evidence, ownership and review.

Concept	Engineering meaning
NIST AI RMF	Risk-management structure that helps organise governance, mapping, measurement and management activities.
TEVV-Athlon	2026 draft framework for structured test, evaluation, verification and validation across multiple AI system types.
ISO/IEC 42001	AI management-system standard for organisational processes, responsibilities and continual improvement.
ISO/IEC 23894	Guidance for identifying, analysing, evaluating, treating and monitoring AI risk.

11.3 The EU AI Act in 2026: Translating Obligations into Technical Work

Engineers often encounter EU AI Act engineering first as a vocabulary term and only later as a design problem. The AI Act applies a risk-based legal structure that engineers must translate into system classification, documentation, transparency, data governance, monitoring and human-oversight work. Article 50 transparency obligations apply from 2 August 2026. Following the 2026 AI Omnibus changes, high-risk rules for Annex III areas apply from 2 December 2027 and rules for high-risk systems embedded in regulated products from 2 August 2028. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a chatbot requirement to inform people they are interacting with AI, or machine-readable marking duties for certain generated content. These are not abstract ethical aspirations; they become interface, metadata, logging and release requirements. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The EU AI Act is the most consequential binding cross-sector AI regulation currently in force. It uses a risk-based structure and imposes different obligations on prohibited practices, high-risk systems, transparency-risk systems, and general-purpose AI. Article 50 transparency obligations apply from 2 August 2026, including duties concerning direct AI interaction and the identification or labeling of certain generated or manipulated content. General-purpose AI obligations began applying earlier, in August 2025.

A critical date correction is necessary because many older summaries are now outdated. Regulation (EU) 2026/1744, the AI Omnibus, moved the core Chapter III high-risk requirements. For systems classified as high risk under Article 6(2) and Annex III, the relevant rules apply from 2 December 2027. For high-risk AI embedded in regulated products under Article 6(1) and Annex I, they apply from 2 August 2028. Engineers should therefore derive requirements from the current consolidated legal text rather than from slides prepared before July 2026.

The most useful governance requirement is one that can be checked. 'Maintain human control' is ambiguous. 'Transfers above a defined value require an authenticated human approval from a separate channel' is executable. 'Protect privacy' is broad. 'No retrieval call may cross a user's tenant boundary, and every access is logged against an authorization decision' can be tested. A mature trustworthy-AI program continually translates policy language into system invariants, tests, logs, and review gates.

This also changes how evidence is designed. Create a regulatory applicability record, map each obligation to a product requirement and evidence artifact, keep provider-versus-deployer responsibilities explicit, and version the mapping as guidance and harmonised standards evolve. Treat legal interpretation as specialist input and technical compliance as an engineering implementation problem. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Regulatory compliance and trustworthiness overlap but are not identical. A compliant system can still be poorly designed for a particular context, and legal requirements evolve. This chapter is engineering orientation, not legal advice. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority.

This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its EU deployment receives a compliance map alongside the assurance case. Transparency notices, generated-content metadata, logs and human-oversight controls are implemented as product features with regression tests rather than appended to a policy document after release. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about the EU AI Act in 2026: translating obligations into technical work is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

We now have the pieces: system boundaries, reliability, uncertainty, explanations, fairness, privacy, security, evaluation, generative AI, agents, people and governance. The final chapter assembles them into a single assurance discipline and asks what trustworthy AI could become beyond today's collection of partially connected practices.

12. Assurance Engineering and Beyond the State of the Art

Engineering question: Can trustworthy AI become a continuously executable engineering discipline rather than a periodic collection of reports?

The mature form of trustworthy AI resembles safety and security engineering more than a checklist of ethical properties. Claims have owners. Evidence is versioned. Tests are reproducible. Counter-evidence is retained. Changes invalidate specific arguments. Monitoring updates confidence after deployment. Meridian's evolution has made this architecture visible one layer at a time. The frontier question is whether these practices can be made compositional and partially executable so that assurance scales with systems that change faster than human review cycles.

12.1 Building an Assurance Case

A course on building an AI assurance case should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. A practical assurance case begins with a bounded top-level claim, decomposes it into subclaims and attaches evidence whose scope is explicit. It should include defeaters: observations that would weaken or invalidate the argument. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a claim that “Meridian is appropriate for internal policy research under approved repositories and read-only tools.” Subclaims cover access control, factual grounding, reliability under document shift, injection containment, human escalation and monitoring. No single benchmark can establish the parent claim. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

The first document should describe the decision or action the system affects. Who is affected? What can go wrong? Which harms are unacceptable? What is the maximum tolerable residual risk? What parts of the workflow can be reversed? Which users are expected to detect errors? This produces the context needed to select technical tests. Beginning with the model encourages teams to test what is easy rather than what is relevant.

Each sentence above should become a set of measurable or reviewable subclaims. The word sufficiently forces a threshold decision. The phrase under expected conditions forces an operating envelope. The phrase known shifts and attacks forces a threat model. The phrase trigger re-evaluation forces lifecycle integration. This is why assurance cases improve thinking even before they improve documentation.

Low-consequence systems may be adequately justified by retrospective evaluation, red teaming, and monitoring. High-consequence systems may require external datasets, prospective trials,

formal verification of critical controls, human-factors studies, independent audit, staged deployment, and strong post-market monitoring. The MASAI mammography trial is instructive because randomized prospective evidence about workflow and detection is fundamentally stronger than a retrospective accuracy table. The evidence burden should scale with exposure and irreversibility.

This also changes how evidence is designed. Maintain a claim-evidence graph, link test results to exact versions, rate evidence independence and freshness, record unresolved hazards and define acceptance authority. Review the argument from the perspective of a skeptical engineer who is trying to falsify it, not a project team trying to complete a template. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. An assurance case can become bureaucratic if claims are vague or evidence is copied without causal relevance. Structured argument is valuable only when it changes design, testing or release decisions. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

An assurance case has a configuration scope. A new base model, a changed system prompt, a new embedding model, a modified retrieval corpus, a new tool, a new permission, or a new user population can invalidate part of the evidence. Change-impact analysis should map each change to the claims it can affect and the tests that must be rerun. This is the bridge from static compliance to continuous assurance.

Trustworthy engineering benefits from remembering failures. A benchmark where the system performed badly, an attack that succeeded, or a subgroup with uncertain results should not disappear after a mitigation is applied. It becomes part of the regression suite and part of the rationale for the control. An organization that preserves only the final passing report throws away the most useful knowledge it generated.

Consider an enterprise research assistant that reads an approved document collection and drafts technical answers with citations. The top-level claim might be that the system is acceptable for internal research support but not for autonomous external publication. The factuality subclaim requires that substantive claims be linked to retrievable source passages. The confidentiality subclaim requires tenant-scoped retrieval and no cross-user memory leakage. The reliability subclaim requires that retrieval or model failure results in an explicit inability to answer rather than invented evidence. The human-oversight subclaim requires review before externally consequential use.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its release review reads like an engineering argument: what the system is allowed to do, why the team believes the controls work, which failures were observed, what remains uncertain and who accepts that residual uncertainty. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about building an assurance case is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Continuous assurance keeps the argument alive



Figure 12. Continuous assurance refreshes evidence after system change.

12.2 Continuous, Executable, and Compositional Assurance

A course on continuous assurance should begin with the failure it is meant to prevent, not with the metric that happens to be fashionable. Static documents decay faster than modern AI systems. Continuous assurance treats evidence as living state: code changes, model updates, policy modifications, drift and incidents automatically identify which claims must be reconsidered. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a dependency update causing a retrieval ranking change. Instead of waiting for an annual audit, the assurance graph knows that citation-grounding evidence depends on retrieval behavior and schedules the affected regression tests before promotion. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Static certification fits artifacts that change slowly. AI systems may change weekly or continuously through model updates, retrieval data, prompts, tool integrations, and user feedback. A future assurance platform should maintain a dependency graph from system components to claims and evidence. When a component changes, it should identify which claims are potentially invalidated, rerun the relevant evaluations, compare results with thresholds, and record the new evidence. This would turn assurance from a document produced before release into a living computational object.

Natural-language policy is flexible but ambiguous. Purely symbolic policy is precise but brittle. A promising architecture is to use language models for semantic interpretation while keeping critical constraints in executable representations. The model can understand an instruction, propose a plan, or map a document to structured facts; a rule engine, type system, model checker, theorem prover, or policy monitor can then enforce constraints that should not depend on probabilistic compliance. This is not a claim that neurosymbolic AI solves alignment. It is a way to allocate tasks according to the strengths of different computational mechanisms.

Examples include an LLM translating a user's request into a typed action plan while a deterministic authorization layer checks permissions; a scientific agent extracting claims while a provenance engine requires source spans; or a document-review system generating conclusions only after a symbolic consistency checker confirms that mandatory evidence relationships are present. The interesting research question is how to make these boundaries expressive enough for real work without reintroducing hidden informal logic inside code.

This also changes how evidence is designed. Express assurance dependencies in machine-readable form, collect evidence from CI/CD and monitoring, preserve immutable provenance, define invalidation rules and require human review only where judgment is genuinely needed. Composition should be cautious: a trustworthy model plus a trustworthy tool does not automatically yield a trustworthy agent unless their interface assumptions are compatible. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is

often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. Executable assurance can automate bookkeeping and some checks but cannot automate all value judgments, threat discovery or scientific interpretation. A formal-looking graph can still encode weak assumptions. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

For autonomous systems, some properties are easier to enforce at runtime than to guarantee during training. A runtime monitor can reject actions outside an approved state space, detect rate or resource anomalies, require human confirmation, or switch to a safe controller. Classical runtime assurance separates a high-performance controller from a verified safety mechanism. An analogous pattern for AI agents would allow a capable probabilistic planner to operate only inside an externally enforced action envelope.

LLMs can already help instantiate structured assurance-case patterns, and this creates a useful division of labor. AI can generate candidate claims, map test results to requirements, identify missing evidence, and update documentation after changes. But an automatically generated assurance case is not assurance unless the evidence is independently checkable. The strongest future systems will use AI to maintain the structure while programmatic tests, formal tools, provenance systems, human reviews, and external audits anchor the claims.

The most ambitious synthesis is an evidence operating system for AI. Every important system artifact would be versioned. Every claim would link to the tests and assumptions that support it. Every production observation could update the evidence. Every model or policy change would trigger an impact analysis. Every incident would become a new regression case. Formal invariants would be checked continuously where possible, while empirical evaluations would sample the open-ended properties that cannot be formalized. Human reviewers would inspect the cases with the highest uncertainty or consequence rather than manually redoing everything.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its assurance ledger becomes part of the deployment pipeline. A new connector cannot be promoted until required evidence is present, while a production incident automatically marks related claims as disputed and reduces authority until review. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The

same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

Continuous assurance keeps the argument alive



Figure 13. Continuous assurance refreshes evidence after system change.

12.3 Beyond 2026: Mechanistic Assurance, Evaluation-Resistant Evaluation, and Scalable Oversight

Engineers often encounter frontier trustworthy AI first as a vocabulary term and only later as a design problem. Several research directions could change assurance rather than merely add another metric. Mechanistic assurance seeks causal evidence about internal computation; evaluation-resistant evaluation studies how to test systems that may recognise or adapt to tests; scalable oversight asks how weaker or resource-limited supervisors can evaluate stronger systems; AI-for-assurance uses models to generate tests, inspect traces and search for counterexamples. This distinction matters because the same numerical result can support very different conclusions depending on the deployment context. The relevant question is always conditional: what exactly has been observed, under which assumptions, and which future decision is that evidence supposed to justify? Keeping those three parts together prevents a familiar failure of AI projects, in which a technically correct measurement is gradually retold as a much stronger claim than the experiment ever established.

Consider a future coding agent that can redesign its own workflow faster than reviewers can read the code. Traditional sample-based review may not scale, so evidence could combine verified action boundaries, adversarially generated tests, interpretability signals, independent models and continuous runtime monitors. What makes the example useful is the separation between mechanism and symptom. The visible symptom may be an incorrect answer, an unfair alert, an unsafe action or a privacy leak, but the mechanism can sit in the data pipeline, model, retrieval layer, policy engine, user interface or organisational process. A trustworthy-AI investigation therefore resists the urge to repair the first component that looks suspicious. It reconstructs the path that produced the outcome and asks which intervention would actually break that path.

Mechanistic interpretability could become more valuable if it is tied to concrete assurance questions. Instead of asking whether a model contains an interpretable feature, ask whether a discovered mechanism predicts a failure mode, whether the mechanism is stable across model versions, whether an intervention reliably changes the relevant behavior, and whether that result

helps bound risk. The 2026 work on sparse-autoencoder consistency points toward this shift from illustrative features to reproducible measurement.

As models are trained on public benchmarks and optimized against automated judges, evaluation itself becomes part of the adaptive environment. Future trustworthy-AI evaluation will need private or continuously refreshed test distributions, adversarial generation of counterexamples, cross-model judging, human expert spot checks, contamination analysis, and metrics designed around deployment invariants rather than fashionable benchmark tasks. The objective is not secrecy for its own sake. It is to preserve the independence between the system being optimized and the measurement used to validate it.

A long-term challenge appears when AI systems become better than individual human evaluators at the tasks they perform. Research on debate, weak-to-strong generalization, recursive evaluation, and related forms of scalable oversight asks whether weaker supervisors can extract reliable evidence from stronger systems. Results remain preliminary. The promising idea is to change the supervision task: instead of requiring a human to solve the whole problem, let the system produce decompositions, adversarial critiques, evidence, or competing arguments that make local verification easier. The risk is circularity if all evidence ultimately comes from similarly flawed models.

This also changes how evidence is designed. Invest in heterogeneous evidence rather than one promised silver bullet. Encourage independent evaluators, hidden and renewable tests, causal interventions, formal constraints around critical actions, automated search for failures, and evidence systems that preserve provenance. Research should measure when these methods fail as carefully as when they succeed. Evidence should be collected at the level at which a claim can be falsified. Averages are helpful for orientation, but engineers need slices, counterexamples, traces, thresholds and failure cases when those are what determine whether authority should be granted. The most valuable evaluation is often not the one with the most samples; it is the one that can cause the team to change a release decision because it directly exercises an assumption on which the system depends.

A useful discipline is to separate capability evidence from assurance evidence. Capability evidence asks whether the system can perform a task under some conditions. Assurance evidence asks whether it can be relied upon for a specified purpose, which also requires information about failure behavior, detection, containment, human response and change over time. The distinction becomes increasingly important for general-purpose models because impressive competence on difficult benchmarks can coexist with surprisingly simple failures. A trustworthy system needs an account of both sides of that jagged performance profile.

The boundary of the claim must remain visible. None of these directions currently provides a general solution to alignment or trustworthy autonomy. Mechanistic interpretability remains incomplete, scalable oversight is an open problem, evaluators can share model biases, and more automation can create correlated assurance failures. This is not pedantry. Learned systems are deployed in open environments, and evidence can expire when data, users, models, prompts, tools or incentives change. Good engineering documentation therefore says not only what passed, but what was not tested, what assumptions make the result meaningful, what observations would contradict it and what change would force the evidence to be collected again.

Progress in trustworthy AI should eventually be measured by a reduction in unjustified confidence. Better systems will not merely fail less often. They will make their operating envelope clearer, detect more of their own boundary violations, expose stronger evidence to users and auditors, constrain irreversible actions, preserve provenance, and recover more gracefully when assumptions fail. The mature outcome is not an AI that asks society to trust it. It is an AI system whose important claims can be challenged and whose evidence survives the challenge.

There is also an organisational dimension. A control that exists in a design document but is routinely bypassed is not an effective control; a human approval that receives no useful evidence is not independent oversight; a red-team finding that cannot block a release is not a meaningful safety signal. Trustworthiness becomes real only when measurements are connected to authority. This is why the course repeatedly returns to owners, escalation, rollback, incident review and evidence freshness alongside model metrics.

For Meridian, its “beyond” architecture is intentionally conservative: Meridian may help generate tests and triage evidence, but it cannot certify itself. Independent controls, diverse evaluators and human acceptance remain separate authorities. The aspiration is not a machine that declares itself trustworthy, but an assurance system that makes unjustified confidence increasingly difficult to sustain. The running system is deliberately ordinary rather than futuristic. Its importance lies in how small product decisions accumulate: adding a repository changes privacy assumptions; adding a tool changes the threat model; adding memory changes persistence; adding autonomy changes the cost of an error. The same pattern occurs in production AI systems across domains. Architecture and governance determine whether a model weakness remains a recoverable inconvenience or becomes a consequential failure.

The engineering takeaway is therefore deliberately conservative. Do not ask a method to establish more than it can. Combine heterogeneous evidence when failure mechanisms differ, preserve counter-evidence instead of averaging it away, and make safe fallback part of the normal design rather than an emergency afterthought. Trust is a human attitude; trustworthy engineering is the work of making reliance inspectable, bounded and revisable.

A second way to reason about beyond 2026: mechanistic assurance, evaluation-resistant evaluation, and scalable oversight is through counterfactual deployment review. Imagine that the system fails tomorrow in a way that is costly but entirely plausible. Which log would explain the event? Which test should have anticipated it? Which component could have contained it? Which person had authority to reduce exposure before the incident? Working backwards from those questions exposes assurance gaps that are invisible in a model-centric review. It also turns abstract principles into engineering requirements: observability, provenance, reversibility and clear ownership are not administrative extras; they are mechanisms for learning from uncertainty without allowing every failure to become a crisis.

The strongest teams also distinguish absence of evidence from evidence of absence. If a red team did not discover a failure, that is evidence about the search that was performed, not a proof that the failure is impossible. If a subgroup has too few examples to estimate performance, reporting the aggregate score does not resolve the uncertainty. If a formal verifier proves one invariant, it does not certify properties outside the formal model. This epistemic discipline may look modest, but it is one of the most important differences between trustworthy engineering and assurance theater.

Trustworthy AI is therefore not a destination reached by satisfying a list of properties. It is a disciplined way to keep claims proportional to evidence while systems, users and environments change. The practical achievement is modest but powerful: replacing vague trust with justified, inspectable and revisable reliance.

CLOSING SYNTHESIS

Meridian began as a useful text generator and ended as a small socio-technical system with evidence, permissions, monitors and governance. Nothing in that evolution required a mystical definition of trust. Every step asked a narrower question: what authority is being granted, which failure could matter, what evidence supports reliance, what remains uncertain, and what happens when the evidence stops applying?

That sequence is the practical core of trustworthy AI. Model quality remains essential, but it is only one layer. The deeper discipline is assurance engineering: connecting claims to evidence and connecting evidence to decisions. This approach is compatible with today's standards and regulatory requirements, yet it also points beyond them toward continuously executable assurance, stronger causal understanding, renewable evaluation and scalable oversight.

The frontier will not be solved by a single metric or one perfect verifier. A more plausible path is heterogeneous: better models, smaller verified components around them, independent evidence, adversarial testing, uncertainty-aware orchestration, provenance, runtime containment, human institutions capable of revising their beliefs, and AI systems that help search for their own failure modes without being allowed to certify themselves. The outcome is not certainty. It is a system in which confidence has a visible reason and a visible expiration date.

GLOSSARY

Term	Meaning in this book
Assurance case	A structured, reviewable argument that a bounded system claim is sufficiently supported by evidence for a defined use.
Calibration	Agreement between stated predictive confidence and observed frequency of correctness under specified conditions.
Conformal prediction	A family of methods that can provide distributional coverage guarantees under stated assumptions.
Distribution shift	A change between the data-generating conditions represented in evaluation and those encountered in deployment.
Evidence decay	Loss of relevance of prior assurance evidence as models, data, tools, users, policies or environments change.
Operating envelope	The declared range of users, tasks, inputs, environmental conditions and authority for which evidence is intended to apply.
Residual risk	Risk remaining after controls, including uncertainty that has been explicitly accepted rather than silently ignored.
RAG	Retrieval-augmented generation: a pipeline in which external material is retrieved and supplied to a generative model.
Selective prediction	A design in which a model answers only a subset of cases and abstains or escalates others.
TEVV	Test, Evaluation, Verification and Validation: complementary activities for establishing evidence about an AI system.
Threat model	An explicit account of protected assets, attacker capabilities, attack surfaces and relevant assumptions.
Trust boundary	A boundary across which data, authority or identity must not be accepted without an independent control.

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